

Advancing GeoAI for Disaster Resilience:
From Damage Assessment to Insightful Decision Support

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11.13.2026 @ Proposal Defense, Online

"Seeing, describing, and governing disasters — beyond the boundaries of the screen, let the real world become the stage for intelligence." --Yifan Yang

EAR Lab
Explore & Resolve

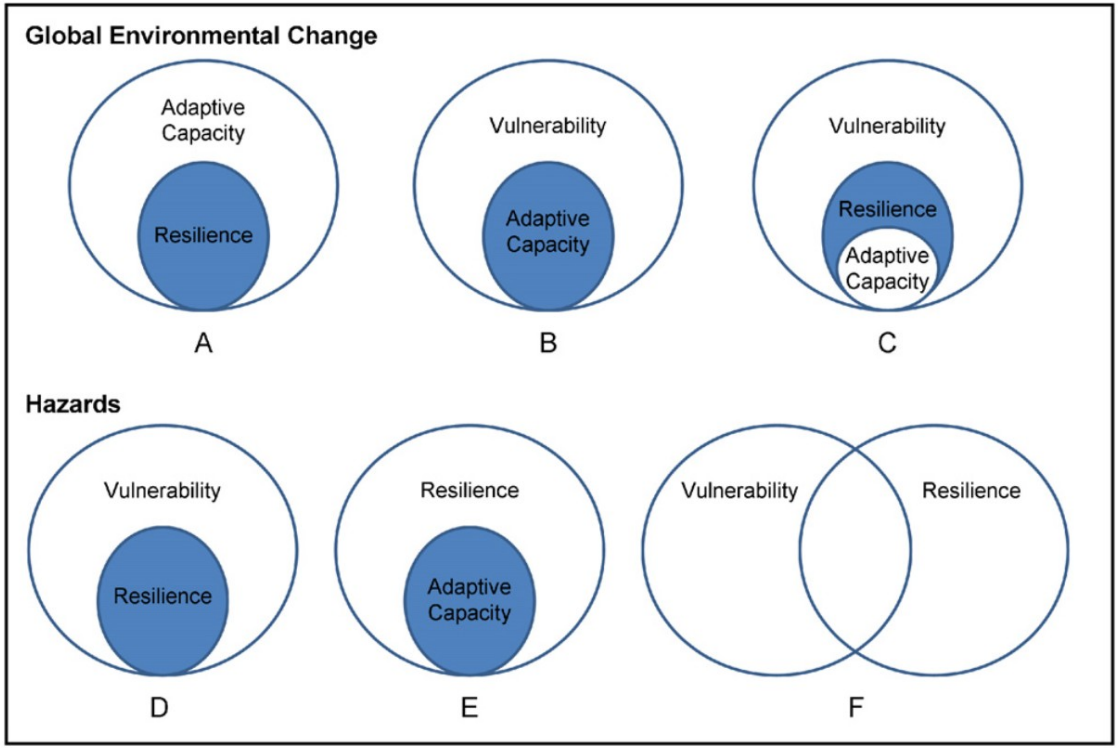
Outline

- Background
- Current Challenges
- Conceptual Framework
- Objectives
- Case Studies
- Publications and Awards
- Research Schedule

Background

Disaster resilience is a geographical question

(1) Rooted in space: resilience is place-based, shaped by local hazard exposure, social vulnerability, and human–environment interactions (Cutter et al., 2008).

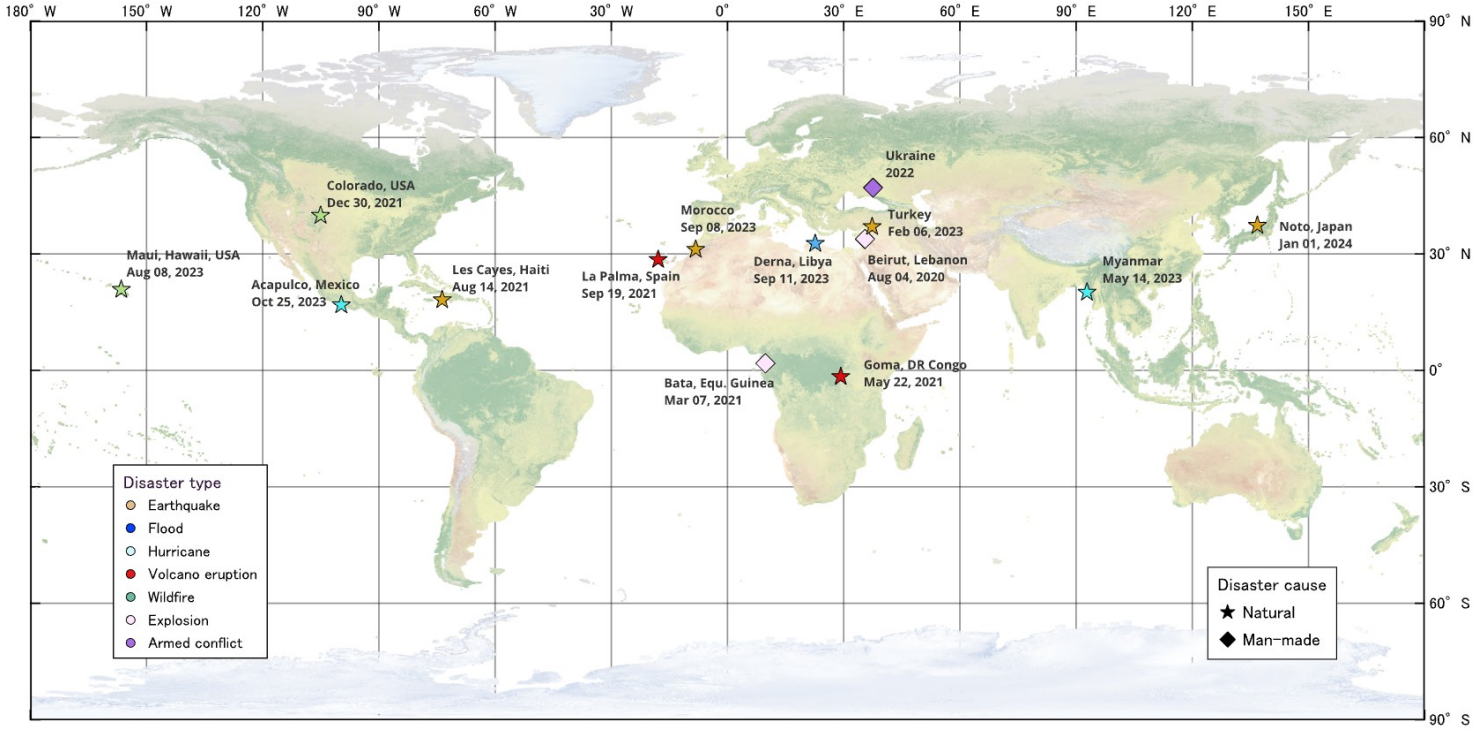


Cutter, S. L., Barnes, L., Berry, M., Burton, C., Evans, E., Tate, E., & Webb, J. (2008). A place-based model for understanding community resilience to natural disasters. Global Environmental Change, 18(4), 598–606.

Background

Disaster resilience is a geographical question

(2) Diverse across the globe: disasters strike different places with different hazards and impacts. The 14 BRIGHT events alone span five continents and seven disaster types, both natural and man-made (Chen et al., 2025).

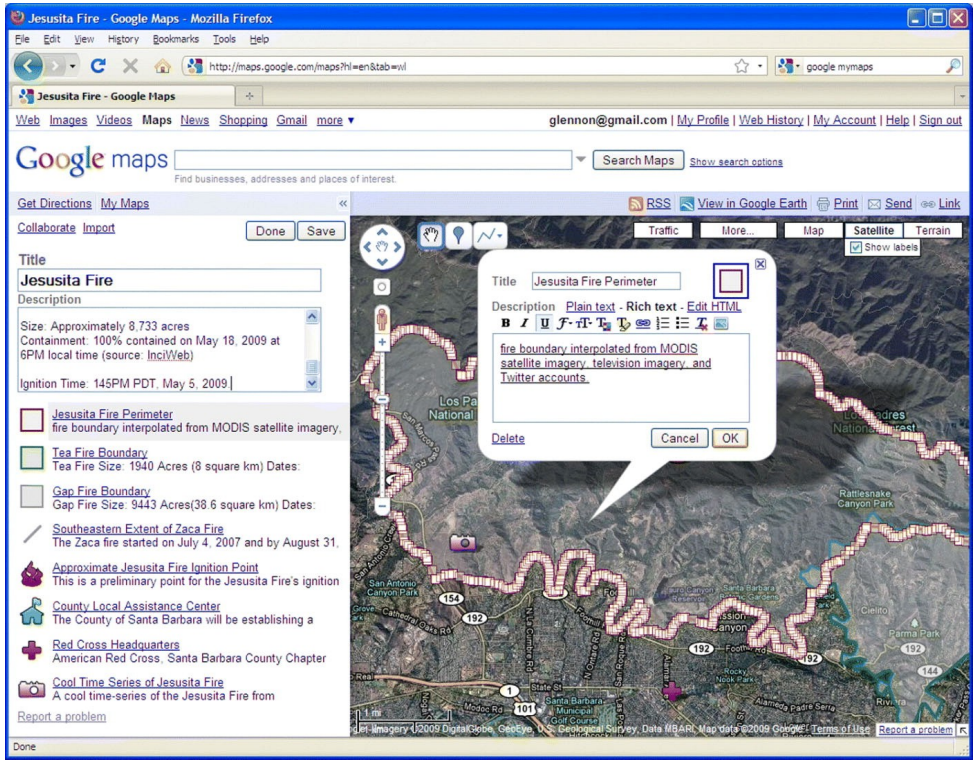


Chen, H., Song, J., Dietrich, O., Broni-Bediako, C., Xuan, W., Wang, J., ... & Yokoya, N. (2025). BRIGHT: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response. Earth System Science Data, 17(11), 6217–6253.

Background

Disaster resilience is a geographical question

(3) A geographic inquiry: beyond what happened, we ask where, how severe, who was affected, and how to allocate resources (Goodchild & Glennon, 2010).



Goodchild, M. F., & Glennon, J. A. (2010). Crowdsourcing geographic information for disaster response: a research frontier. International Journal of Digital Earth, 3(3), 231–241.

Background

Disaster resilience is a geographical question

(4) A cycle in time: resilience is the ability to prepare for, absorb, recover from, and adapt to adverse events (National Academies et al., 2012). Each phase, from preparedness to response, recovery, and mitigation, depends on timely, place-specific information.



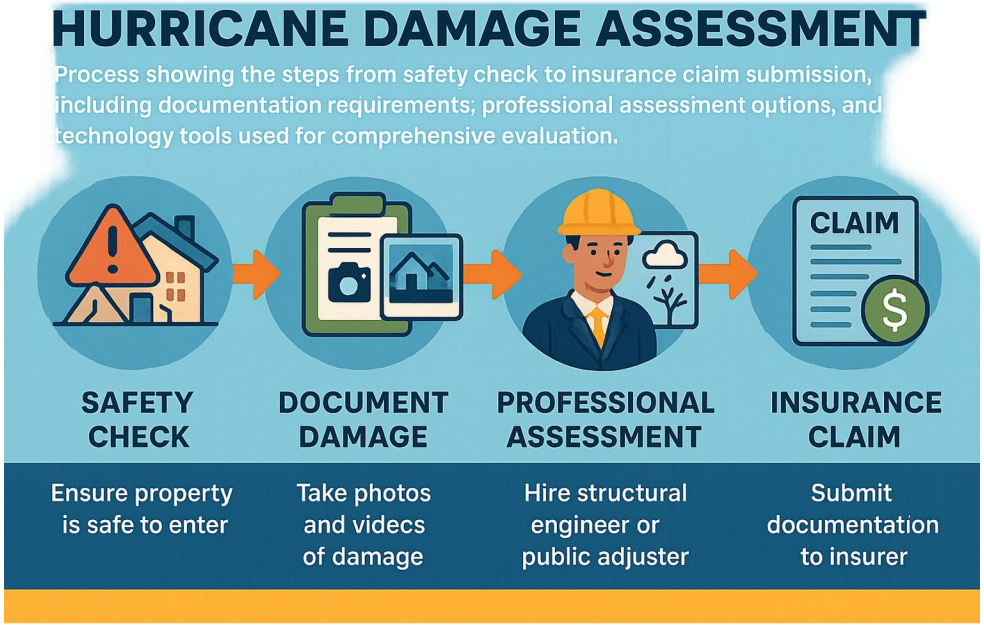
Image generated by ChatGPT

National Academies, Policy, Global Affairs, Committee on Science, Public Policy, & Committee on Increasing National Resilience to Hazards. (2012). Disaster resilience: A national imperative. National Academies Press.

Background

Disaster assessment is a critical task for building resilience

Core tasks: acquire imagery (remote sensing + street view) → classify severity → generate reports → map impacts rapidly (FEMA, 2016).





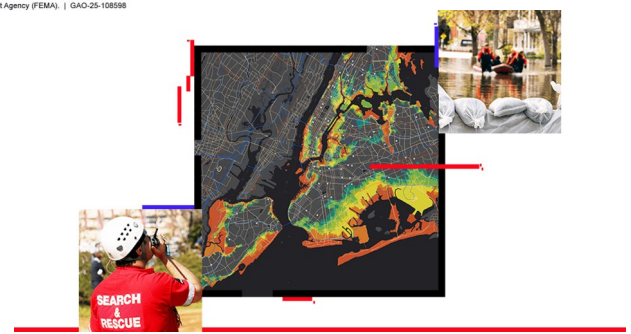
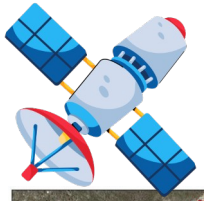


Image source: Insurance Claim Recovery Support, hurricane damage assessment workflow. · FEMA (2016). Damage Assessment Operations Manual: A Guide to Assessing Damage and Impact. U.S. Department of Homeland Security.

Background

Image data is central to disaster assessment: two complementary views



Overhead view — remote sensing

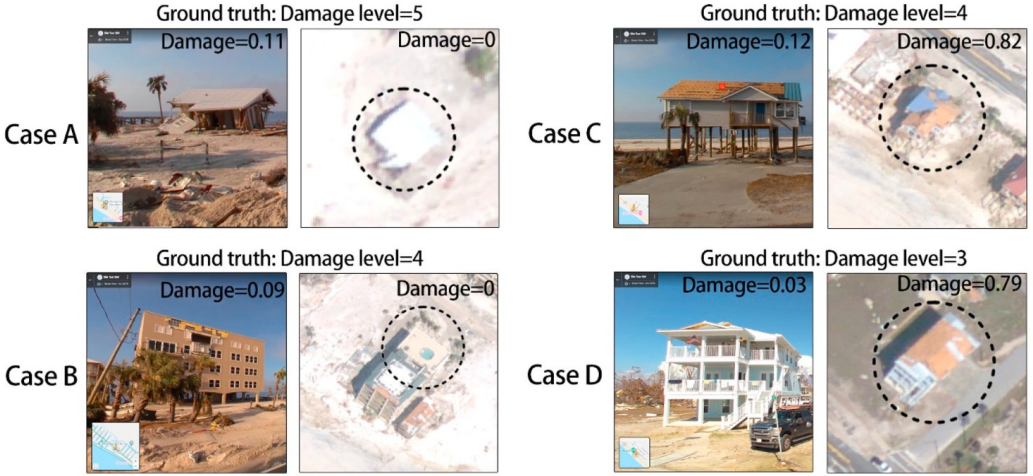
Satellite and aerial imagery: wide coverage and roof-level damage (e.g., xBD benchmark; Gupta et al., 2019).





Ground-level view — street-view imagery

Facade-level detail of structures, debris, and flooding, captured by survey vehicles, panoramic cameras, and volunteers (Zhai & Peng, 2020).



Left: Hauptman, L., Mitsova, D., & Briggs, T. R. (2024). Hurricane Ian damage assessment using aerial imagery and LiDAR: A case study of Estero Island, Florida. Journal of Marine Science and Engineering, 12(4), 668.

Right: Zhai, W., & Peng, Z. (2020). Damage assessment using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. Applied Geography, 123, 102252.

Gupta, R., Hosfelt, R., Sajeev, S., Patel, N., Goodman, B., Doshi, J., Heim, E., Choset, H., & Gaston, M. (2019). xBD: A dataset for assessing building damage from satellite imagery. CVPR Workshops.

Background

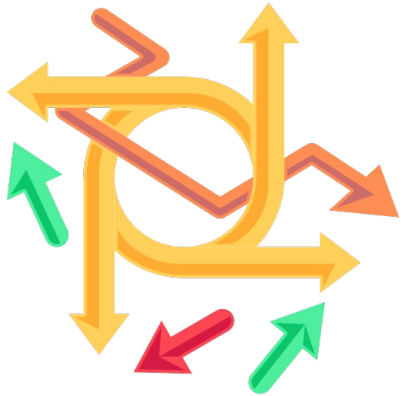
The bottleneck: imagery is abundant, interpretation is not



Volume (too much)



Speed (too slow)

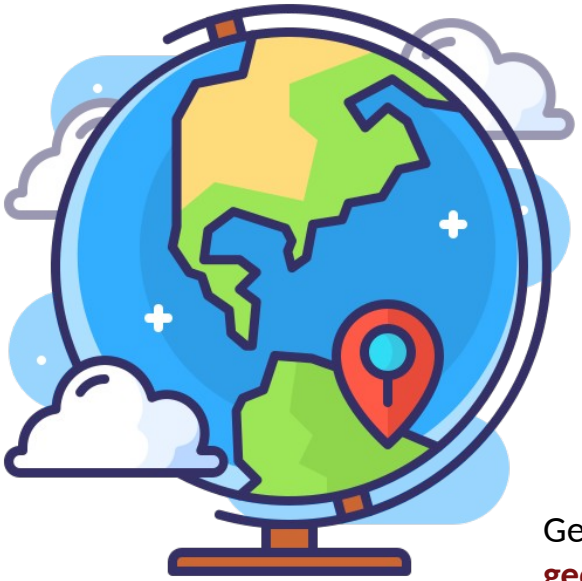


Complexity (too difficult)

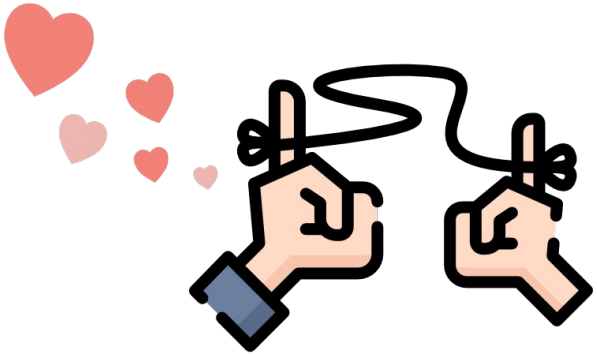
Manual interpretation cannot scale → we need AI that understands geography

Background

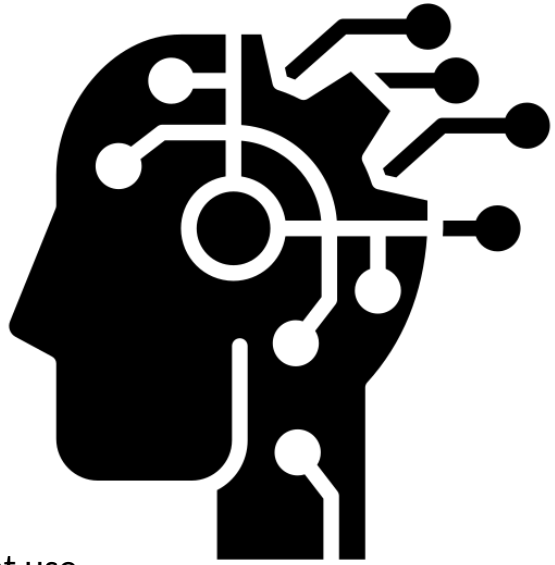
GeoAI for disaster resilience



Geography



Artificial Intelligence



GeoAI refers to **spatially explicit AI** methods that use **geographic data and spatial context** to **create** intelligent geographic information, knowledge, and services.

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. International Journal of Geographical Information Science, 34(4), 625–636.

Background

Three families of AI models, three roles in GeoAI



Discriminative models — task-specific classifiers

"Is this building damaged? Mild, moderate, or severe?"

+ Accurate on the task they are trained for

– Need labels for every event; stay confident even when wrong



Generative models — foundation and vision-language models

"Why is this location severely damaged? What should responders do here?"

+ Zero- and few-shot generalization across hazards and regions

– Usually less accurate than tuned classifiers; may hallucinate



Agentic AI = Large Language Models (LLMs) + tools + context

"Plan the analysis, call the spatial tools, and show the evidence behind every claim."

+ Combines both families: classifiers for accuracy, foundation models for reasoning

+ Calls GIS tools and live data, so answers rest on computed evidence, not memory

+ Extends beyond damage assessment to the whole resilience cycle

Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. arXiv:2108.07258. · Wang, L., et al. (2024). A survey on large language model based autonomous agents. Frontiers of Computer Science, 18(6), 186345. · Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. ICML.

Background — one-sentence summary

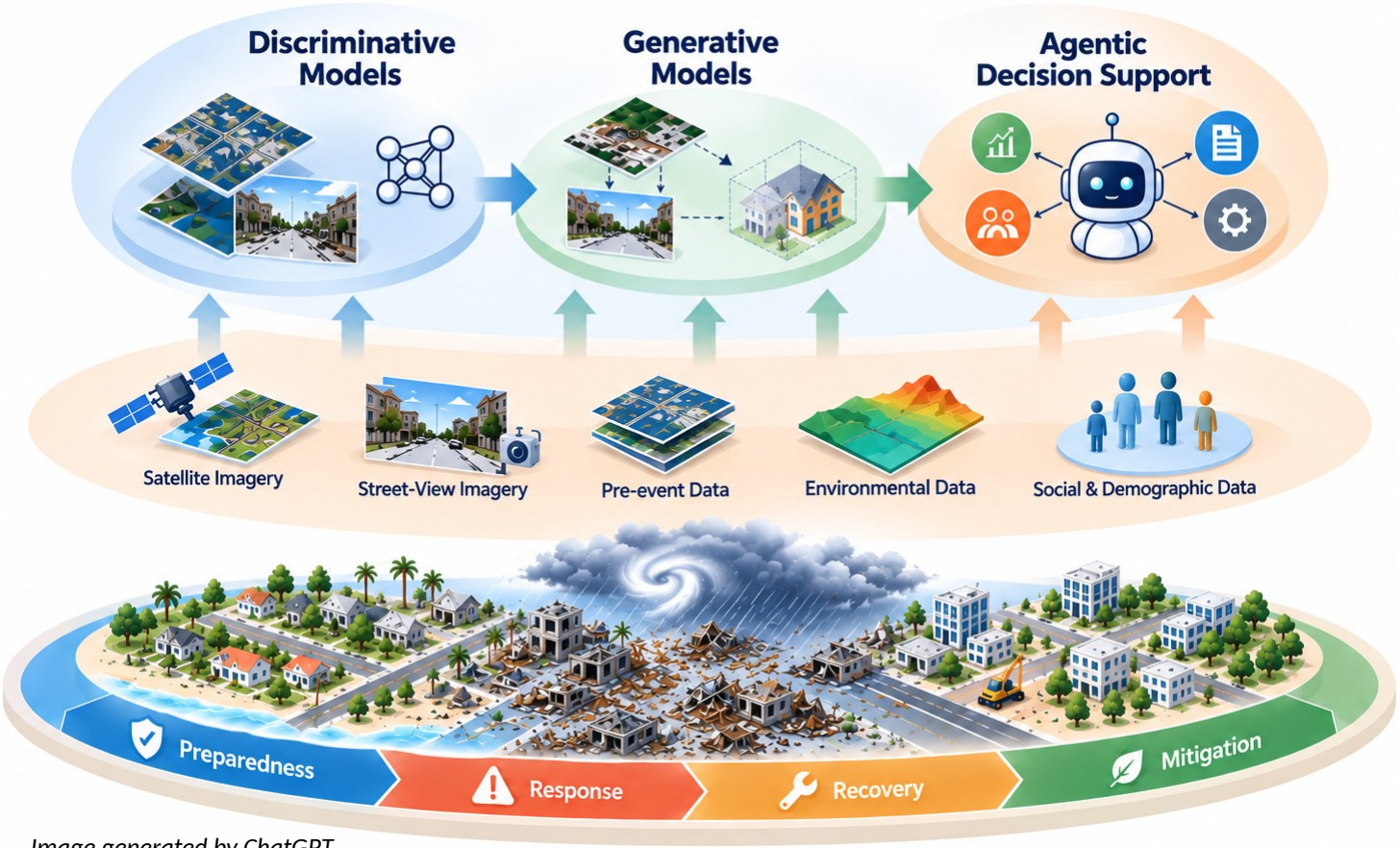


Image generated by ChatGPT

***Advancing GeoAI for Disaster Resilience*: From Damage Assessment to Insightful Decision Support**

Current Challenges

1 **Discriminative assessment is not yet accurate or reliable enough.**

Models that see only one post-event view misclassify damage and stay confident when wrong.

How to see better?

2 **Specialized models do not generalize across events.**

Each new disaster needs new labels and retraining, which is too slow for response.

How to respond faster?

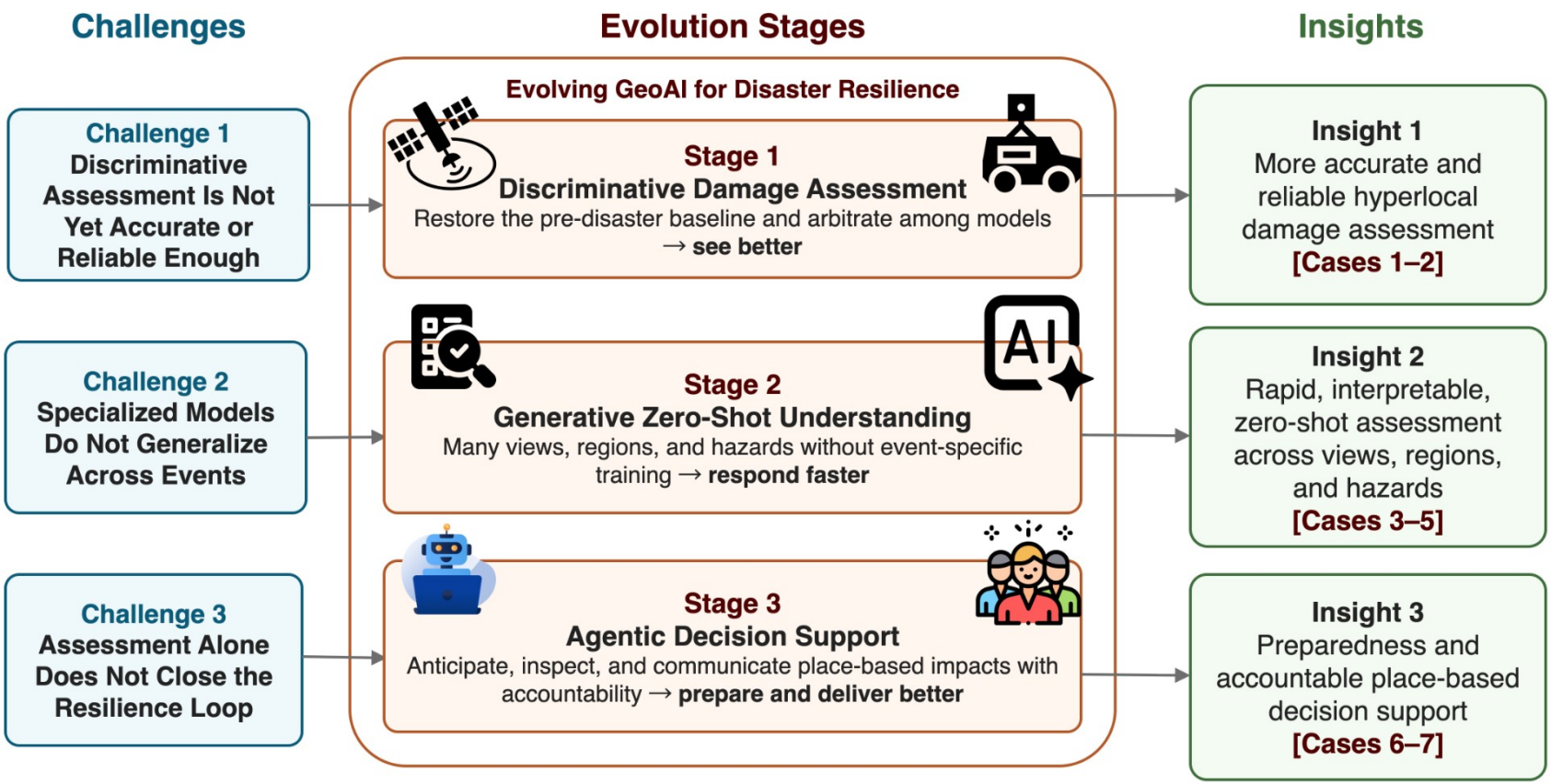
3 **Assessment alone does not close the resilience loop.**

A damage label or map is not a decision; each phase needs actionable, accountable guidance.

How to prepare and deliver better?



Conceptual Framework



Objectives

Objective 1 → *addresses Challenge 1 · Cases 1–2*

Improve accuracy and reliability by fusing different views and arbitrating among multiple models.

See better

Objective 2 → *addresses Challenge 2 · Cases 3–5*

Enable zero-shot assessment that transfers across hazards and regions without new labels or retraining.

Respond faster

Objective 3 → *addresses Challenge 3 · Cases 6–7*

Build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters.

Prepare and deliver better



Since the first defense (October 22): three clarifications

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1 · One thread from question to cases

- Disaster resilience is a geographical question → damage assessment is its hinge → three model families → three challenges, three objectives, seven cases, mapped one to one

2 · Evidence that is easy to read

- Every results table is now a chart with a one-line takeaway; reliability (overconfident errors) is reported next to accuracy, not only accuracy

3 · The ongoing work has a plan

- Cases 6 and 7 now state scope, data, deliverables, evaluation, and milestones through Summer 2027 (slides follow each case)

Subtitle updated: From Damage Assessment to Insightful Decision Support

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STAGE 1 · OBJECTIVE 1

See better

Challenge 1 · Discriminative assessment is not yet accurate or reliable enough

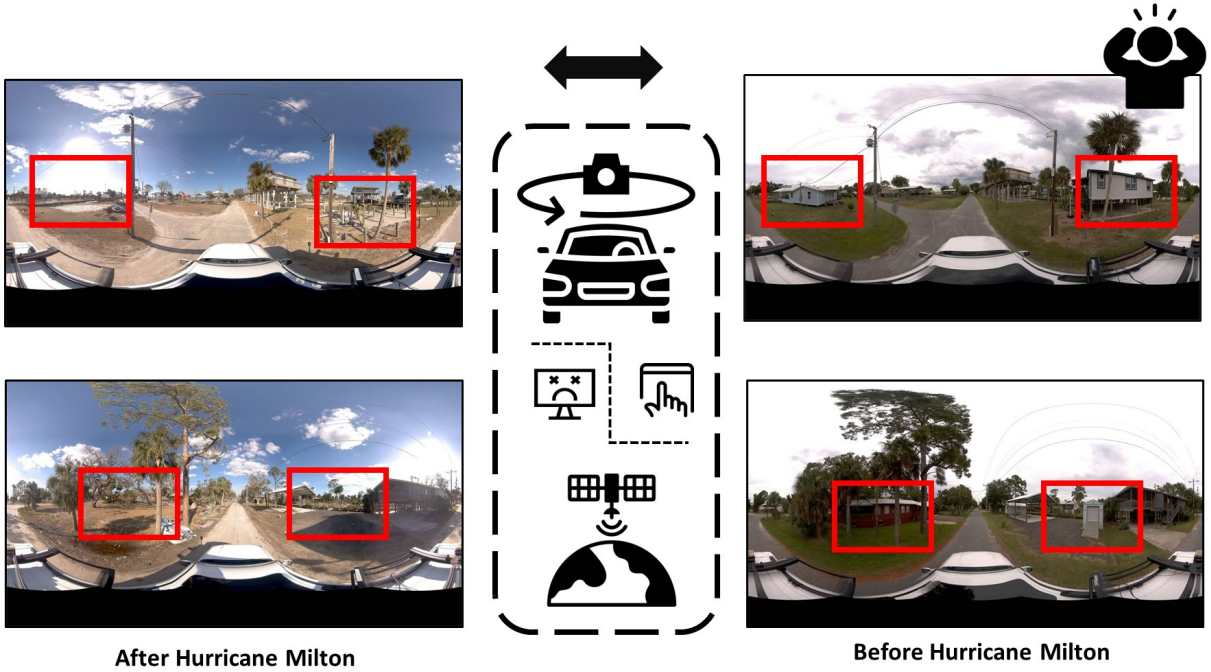
Improve accuracy and reliability by fusing pre- and post-event views and arbitrating among multiple models.

Case 1	**Hyperlocal damage assessment from bi-temporal street-view imagery**	Published · CEUS 2025
Case 2	**DamageArbiter: multimodal arbitration for damage assessment**	Under review · Natural Hazards

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Explore & Resolve

Case 1:

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)



After Hurricane Milton

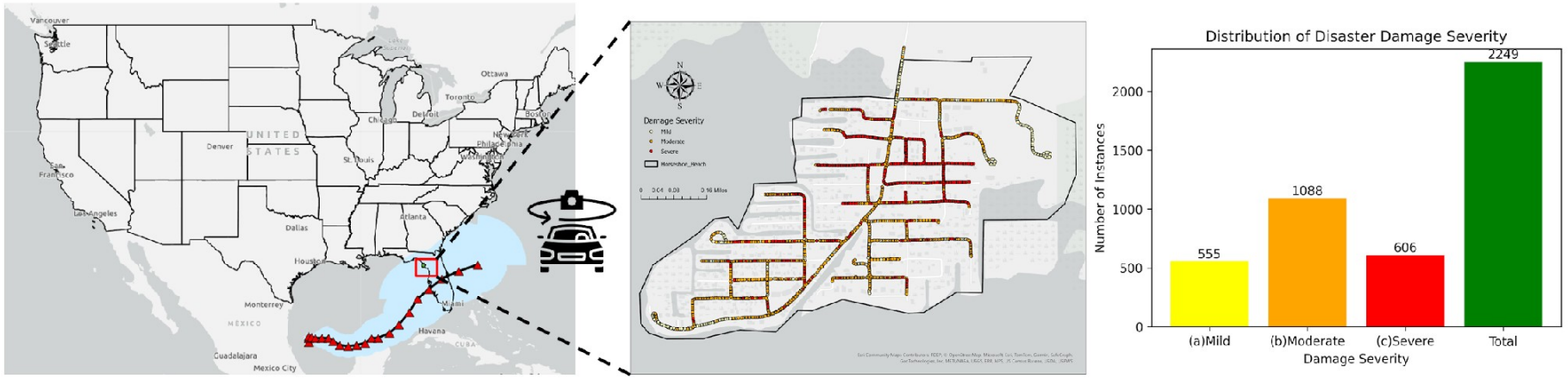
Before Hurricane Milton

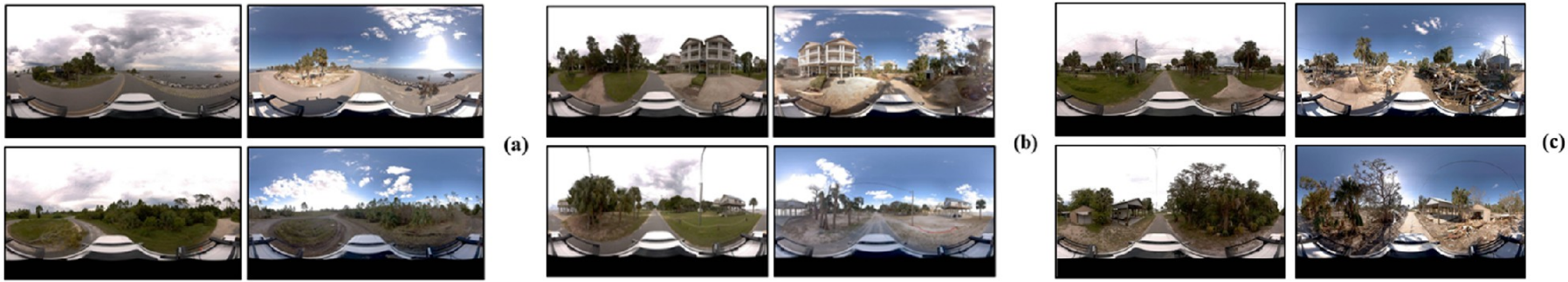
Research questions:

(1) Can pre-disaster street-view imagery help us classify disaster damage more accurately?

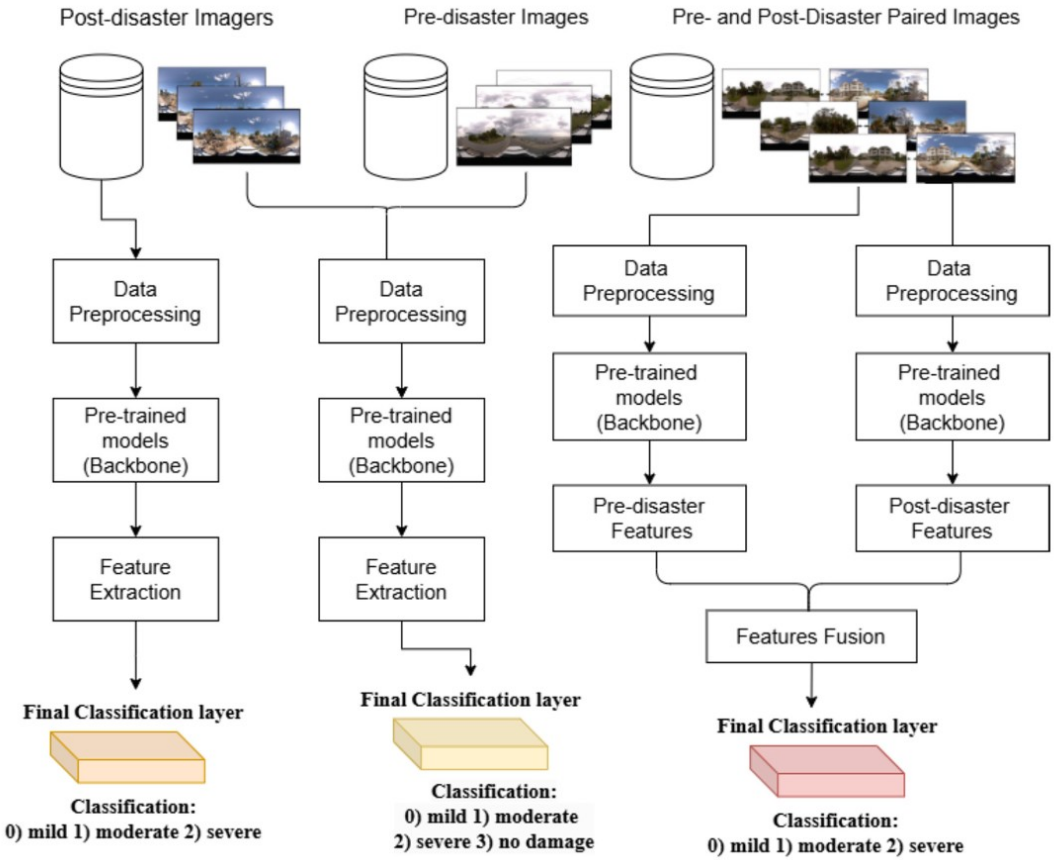
(2) How can we integrate pre- and post-disaster street-view imagery to enable rapid, hyperlocal damage assessment?

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (study area and dataset)



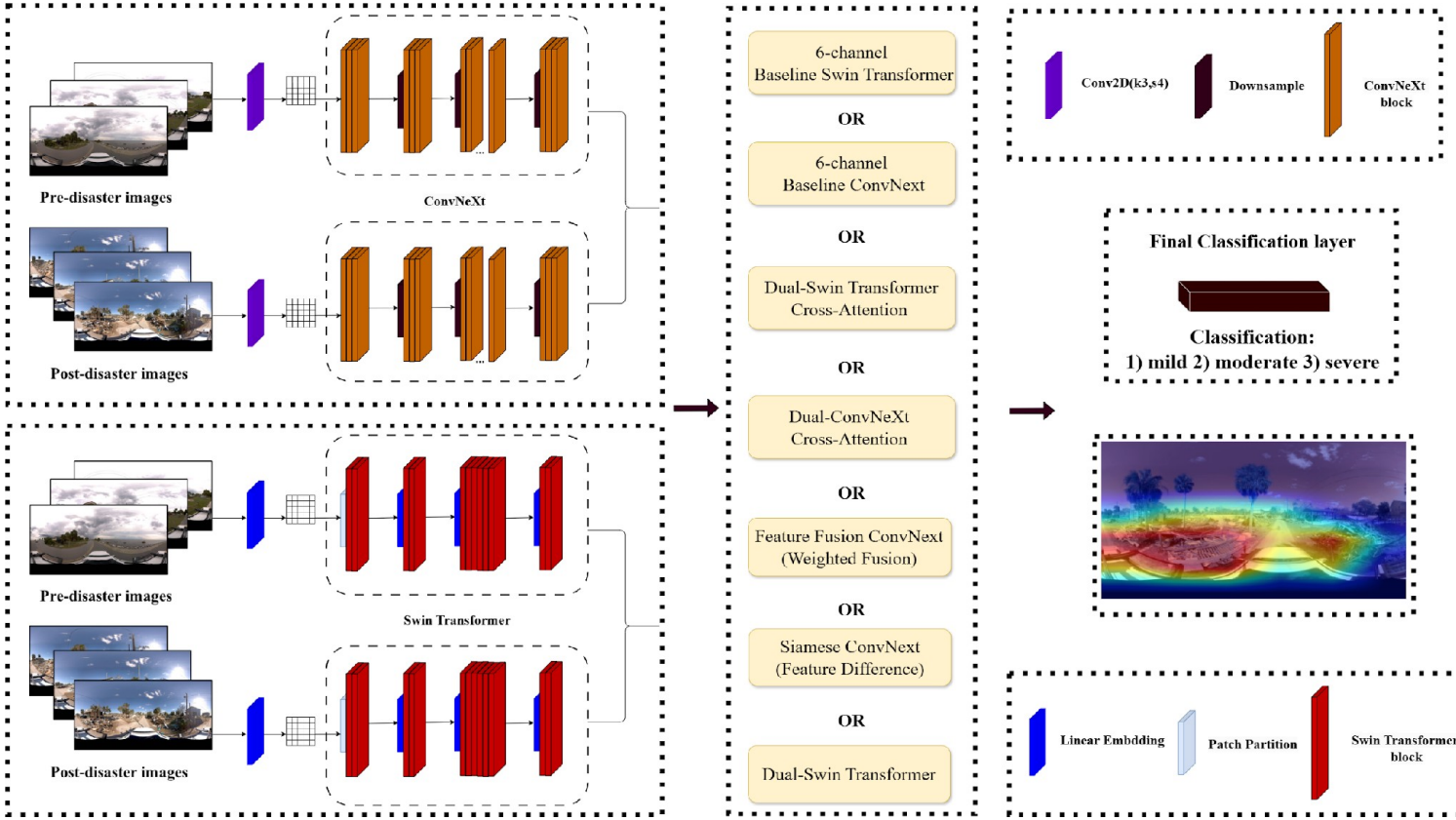


Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

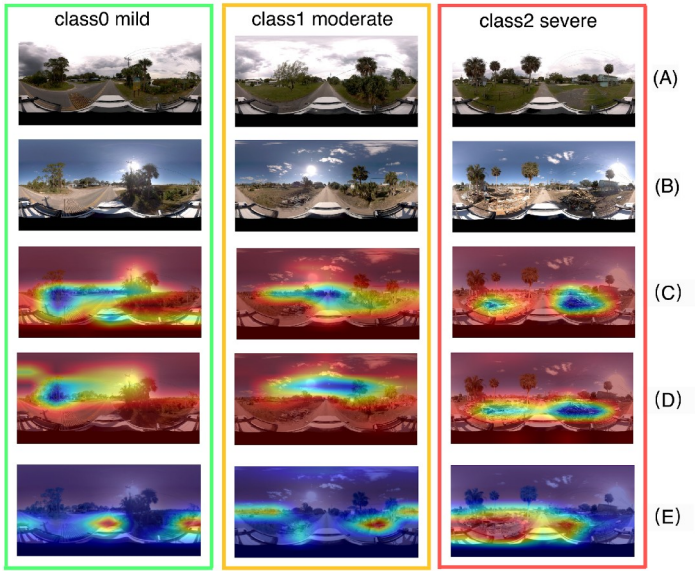


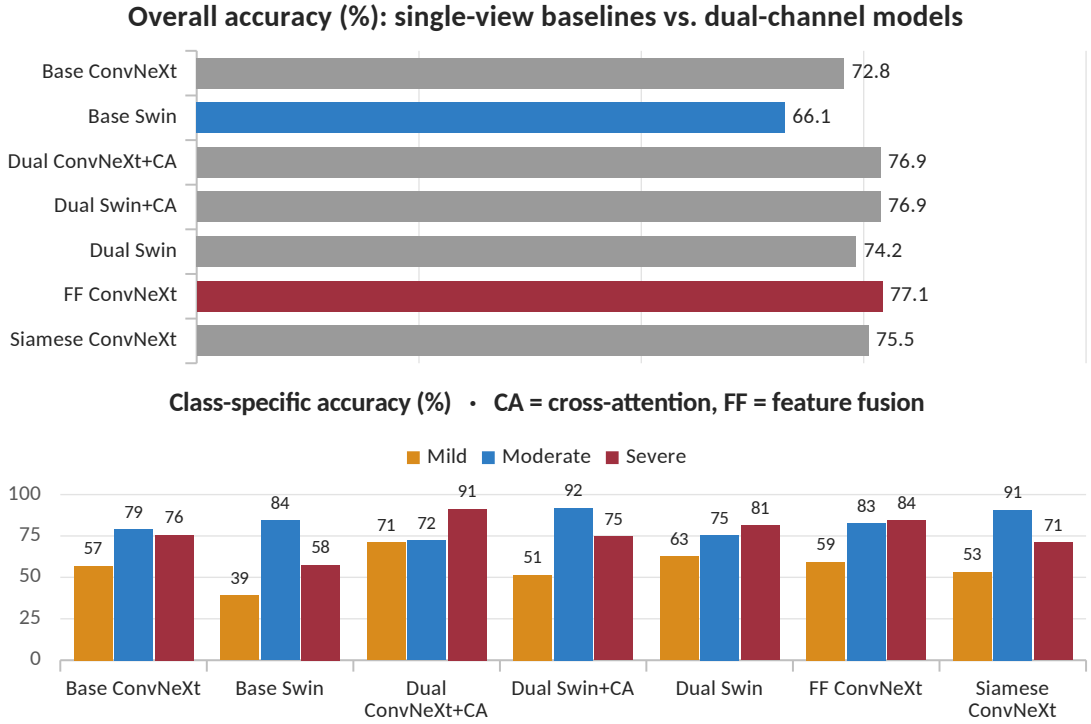
Three settings:
(1) **Post-only**: classify post-event images into 3 levels
(2) **Pre- and Post-**: pre- and post-event images mixed together, 4 classes (+ no damage)
(3) **Pre- and Post-Paired**: pre- and post-event features fused

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

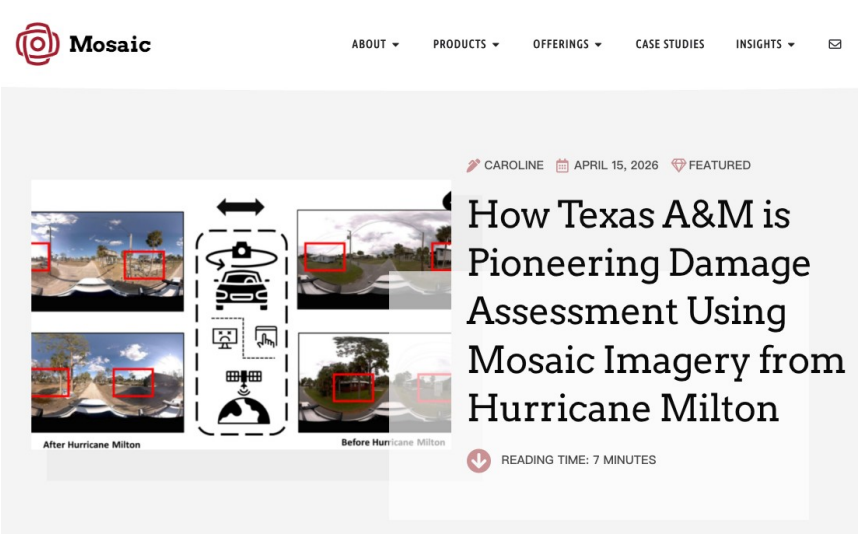




Key result: Accuracy improved from 66.14% (Swin baseline) to 77.11% (dual-channel Feature-Fusion ConvNeXt); Grad-CAM confirms pre-disaster images help focus on true pre/post changes.

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)

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https://www.mosaic51.com/featured/how-texas-am-is-pioneering-damage-assessment-using-mosaic-imagery-from-hurricane-milton/

Computers, Environment and Urban Systems 121 (2025) 102335

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Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

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ARTICLE INFO

Keywords:
Disaster resilience
Street-view imagery
Dual-channel neural network
Pre-trained vision model
Damage estimation

ABSTRACT

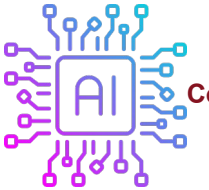
Street-view images offer unique advantages for disaster damage estimation as they capture impacts from a visual perspective and provide detailed, on-the-ground insights. Despite several investigations attempting to analyze street-view images for damage estimation, they mainly focus on using post-disaster images. The potential of time-series street-view images remains underexplored. Pre-disaster images provide valuable benchmarks for accurate damage estimations at building and street levels. These images could also aid annotators in objectively labeling post-disaster impacts, improving the reliability of labeled data sets for model training, and potentially enhancing the model performance in damage evaluation. The goal of this study is to estimate hyperlocal, on-the-ground disaster damages using bi-temporal street-view images and advanced pre-trained vision models. Street-view images before and after 2024 Hurricane Milton in Horseshoe Beach, Florida, were collected for experiments. The objectives are: (1) to assess the performance gains of incorporating pre-disaster street-view images as a no-damage category in fine-tuning pre-trained models, including Swin Transformer and ConvNeXt, for damage level classification; (2) to design and evaluate a dual-channel algorithm that reads pair-wise pre- and post-disaster street-view images for hyperlocal damage assessment. The results indicate that incorporating pre-disaster street-view images and employing a dual-channel processing framework can significantly enhance damage assessment accuracy. The accuracy improves from 66.14 % with the Swin Transformer baseline to 77.11 % with the dual-channel Feature-Fusion ConvNeXt model. Gradient-weighted Class Activation Mapping (Grad-CAM) shows that incorporating pre-disaster images improves the pre-trained vision model's capacity to focus on major changes between pre- and post-disaster images, thus enhancing model performance. This research evaluates the technical solutions and challenges of assessing disaster damages at hyperlocal spatial resolutions with bi-temporal street-view images, providing valuable insights to support effective decision-making in disaster management and resilience planning.

Yang, Y., Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., & Yang, M. (2025). Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models. Computers, Environment and Urban Systems, 121, 102335.

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Natural Hazards)



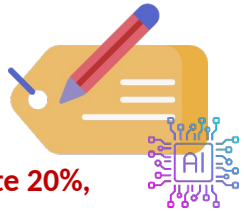
Post-Hurricane Street-View Imagery



Computer Vision Models (e.g., ViT-B/32)

Mild

(mild 70%, moderate 20%, severe 10%)





Actual value: Severe

Overconfidence = models may give very confident predictions even when they are wrong (error = 70% − 10% = 60%)

Motivation

- Accuracy alone is not enough
- Reliability matters for disaster response
- Overconfident wrong predictions
- Risk of misleading prioritization

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

class0 mild



No buildings in the area were damaged by the hurricane, and the trees appear to have sustained only mild damage. This is truly a relief. The vegetation is lush, the trees are tall, and the damage from the hurricane was minimal. I'd love to live here now, as it's only slightly damaged and looks very livable.

Sagging power lines are visible alongside a clear, dry road. Many trees show defoliation and some branch damage, but no large fallen trees or building debris are present. The overall scene suggests mild environmental impact, with infrastructure largely intact.

class1 moderate



The damage level is moderate, with loose building materials and a small amount of debris suggesting recent construction or minor disrepair. Despite the somewhat cluttered appearance of scattered items, the water features and elevated residences make this area appear to be an attractive residential area.

Scattered wooden debris and loose utility lines on a power pole indicate moderate damage along this coastal dirt road. While buildings appear structurally sound, the presence of scattered planks and beams, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

class2 severe

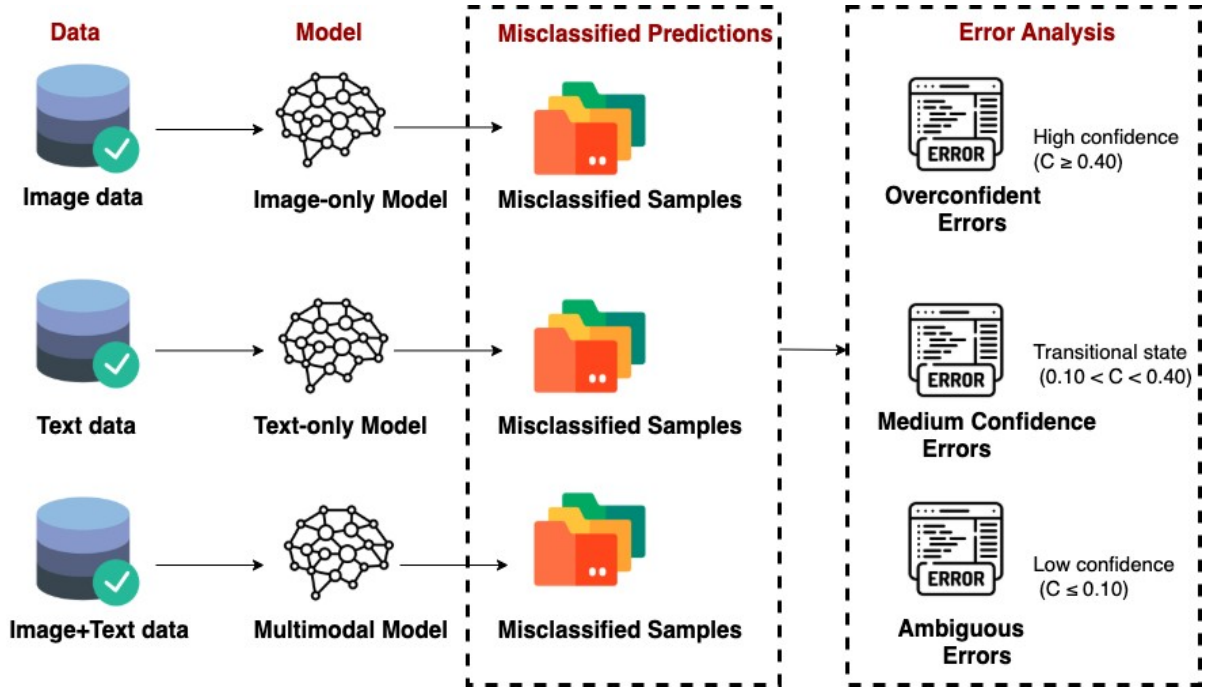


The atmosphere of the image is chaotic, with massive piles of debris covering large parts of the area, giving a strong sense of destruction. The damage appears extensive, with shattered building materials strewn all over, although some elevated houses remain intact in the background.

The street is heavily impacted by extensive building debris, forming large piles along both sides, almost completely blocking passage. This significant accumulation of scattered materials and collapsed fragments suggests severe damage to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

RQ1: How reliable are image-only, text-only, and image-text models for street-view disaster damage assessment?

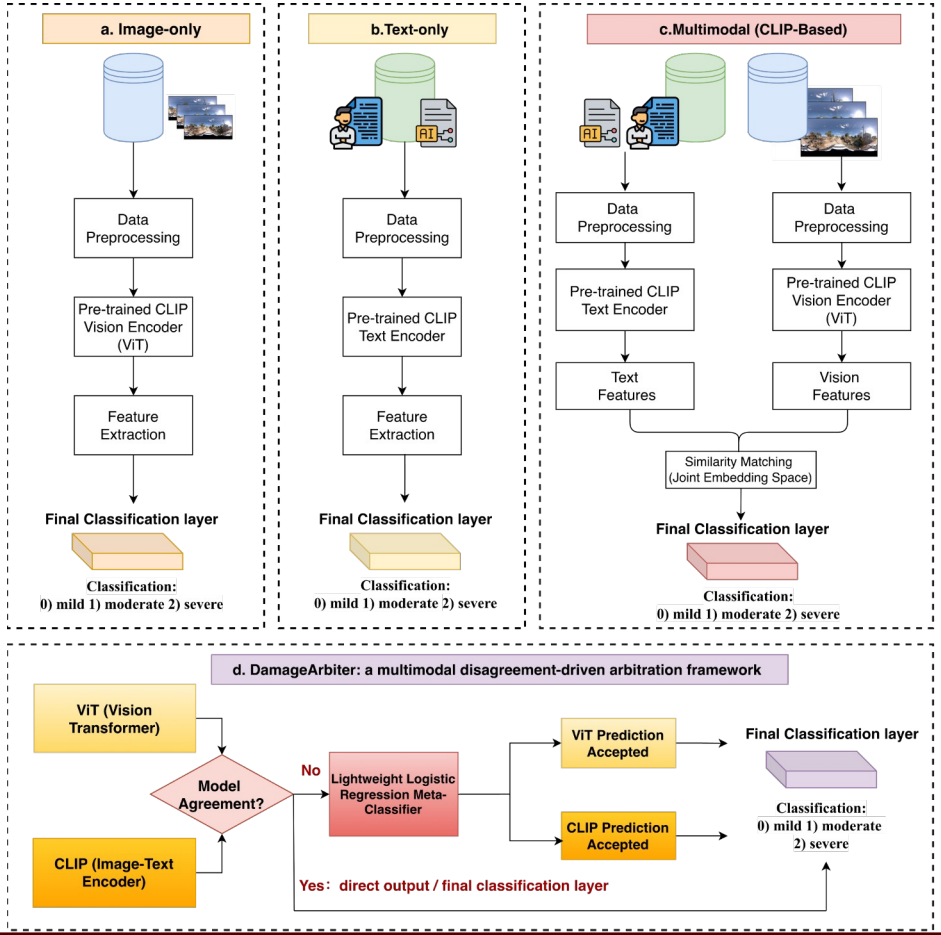
RQ2: Can LLM-generated image descriptions support reliable damage assessment when human annotations are limited?



Overconfidence = models may give very confident predictions even when they are wrong

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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

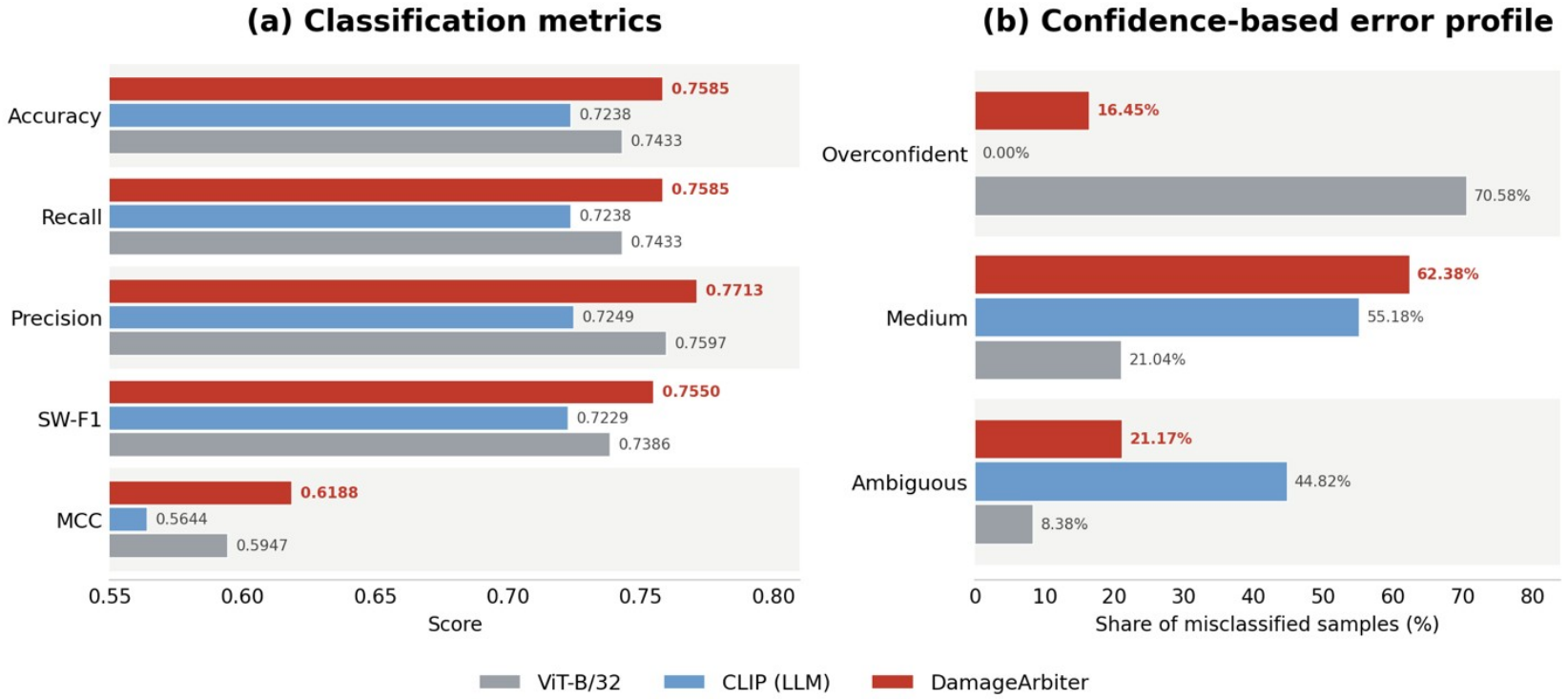


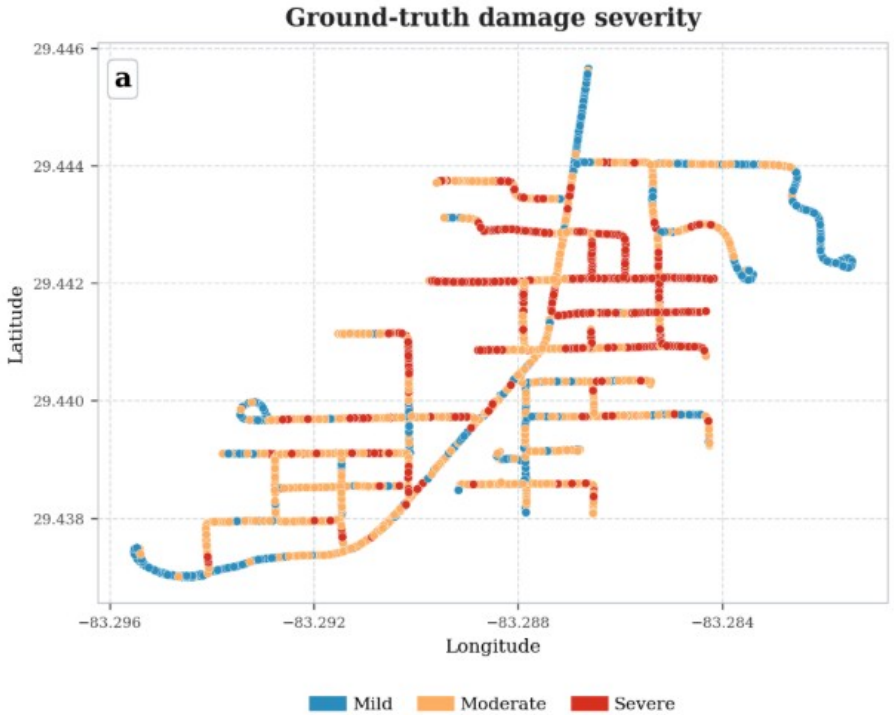
Three single pipelines (a–c)

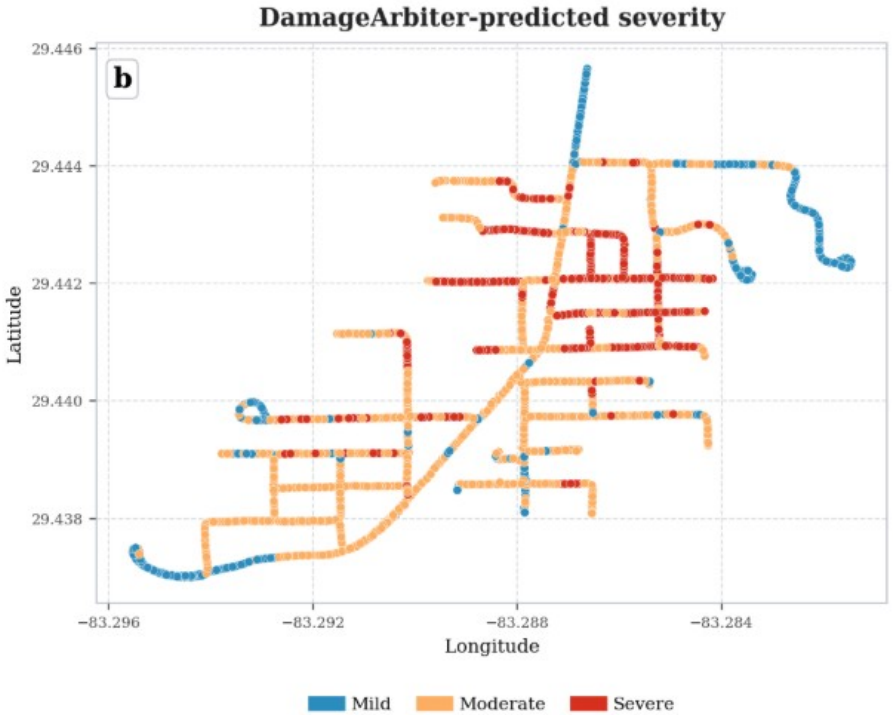
- Image-only: CLIP vision encoder (ViT)
- Text-only: CLIP text encoder on LLM descriptions
- Multimodal: image + text in a joint embedding
- Same output: mild / moderate / severe

Arbitration (d)

- ViT and CLIP predict independently
- Agree → accept directly
- Disagree → a small logistic-regression arbiter picks whom to trust
- Base models untouched; cheap and interpretable

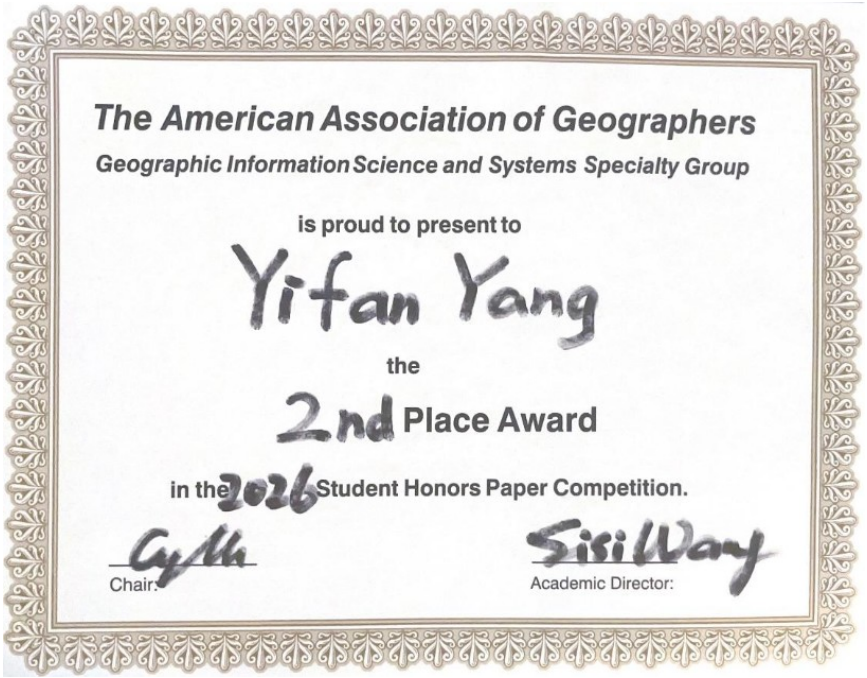

(a) Classification metrics
Accuracy
0.7585
0.7238
0.7433
Recall
0.7585
0.7238
0.7433
Precision
0.7713
0.7249
0.7597
SW-F1
0.7550
0.7229
0.7386
MCC
0.6188
0.5644
0.5947
0.55
0.60
0.65
0.70
0.75
0.80
Score
(b) Confidence-based error profile
Overconfident
16.45%
0.00%
70.58%
Medium
62.38%
55.18%
21.04%
Ambiguous
21.17%
44.82%
8.38%
0
10
20
30
40
50
60
70
80
Share of misclassified samples (%)
ViT-B/32
CLIP (LLM)
DamageArbiter


Ground-truth damage severity
a
29.446
29.444
29.442
29.440
29.438
Latitude
−83.296
−83.292
−83.288
−83.284
Longitude
Mild
Moderate
Severe


DamageArbiter-predicted severity
b
29.446
29.444
29.442
29.440
29.438
Latitude
−83.296
−83.292
−83.288
−83.284
Longitude
Mild
Moderate
Severe

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Natural Hazards)

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The American Association of Geographers

Geographic Information Science and Systems Specialty Group

is proud to present to

Yifan Yang

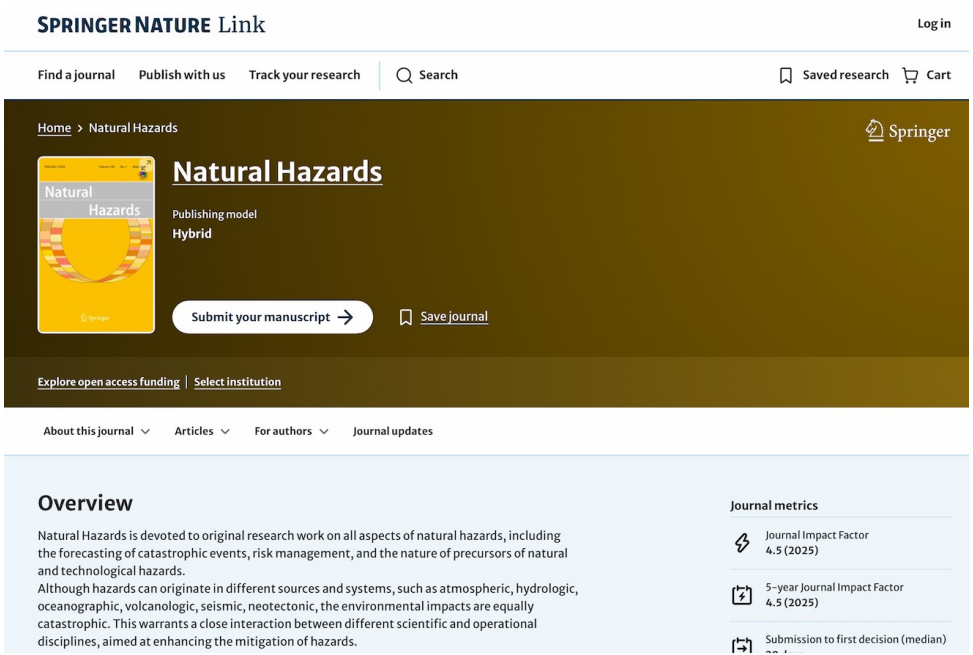
the

2nd Place Award

in the 2026 Student Honors Paper Competition.

Chair:

Academic Director: Sisi Wang



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Overview

Natural Hazards is devoted to original research work on all aspects of natural hazards, including the forecasting of catastrophic events, risk management, and the nature of precursors of natural and technological hazards.
Although hazards can originate in different sources and systems, such as atmospheric, hydrologic, oceanographic, volcanologic, seismic, neotectonic, the environmental impacts are equally catastrophic. This warrants a close interaction between different scientific and operational disciplines, aimed at enhancing the mitigation of hazards.

Journal metrics

Journal Impact Factor
4.5 (2025)

5-year Journal Impact Factor
4.5 (2025)

Submission to first decision (median)

Yang, Y., Zou, L., Gong, W., Fu, K., Li, Z., Wang, S., ... & Tian, H. (2026). DamageArbiter: A CLIP-enhanced multi-modal arbitration framework for hurricane damage assessment from street-view imagery. arXiv preprint arXiv:2603.14837.

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STAGE 2 · OBJECTIVE 2

Respond faster

Challenge 2 · Specialized models do not generalize across events

Enable zero-shot assessment that transfers across hazards and regions without new labels or retraining.

Case 3	**RAPID: multi-agent pipeline for interpretable cross-view assessment**	Accepted · ACM SIGSPATIAL 2026
Case 4	Satellite-to-Street: post-disaster street views from satellite imagery	Accepted · IEEE IGARSS 2026
Case 5	Generative AI for disaster resilience	Under review · CRC Press

Focus today: Case 3 (Cases 4–5 and Case 8 in backup)

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Case 3:
RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted by ACM SIGSPATIAL)



Research Question

Can RAPID support:

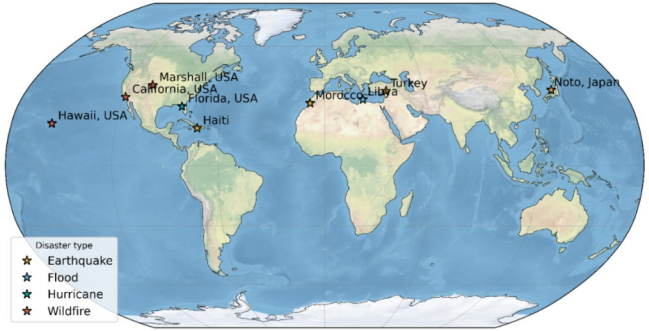
- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?



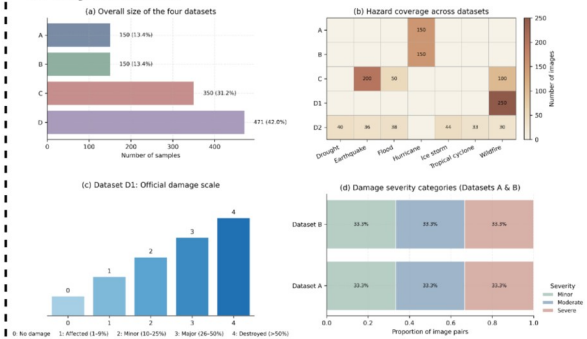
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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a. Geographic location of events

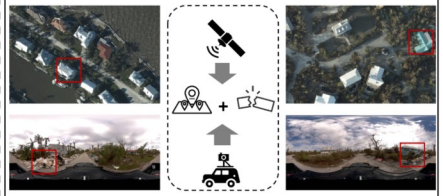


b. Key statistics



c. Dataset description

Dataset A:
Post-disaster street-view and remote-sensing image pairs



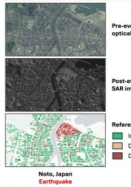
It sourced from the CVDisaster dataset.
It provides paired street-view and remote-sensing imagery from hurricane-affected regions, each annotated with three levels of structural damage (light, medium, heavy).

Dataset B:
Pre- and post-disaster street-view image pairs



It consists of bitemporal street-view image pairs from the BiTemporal StreetView Damage dataset. Each pair includes precise location metadata and corresponding severity labels (minor, moderate, severe), mainly collected from Hurricane-impacted area.

Dataset C:
Pre- and post-disaster remote-sensing image pairs



It (BRIGHT) provides pre-/post-disaster satellite imagery with pixel-level building-damage masks, covering multiple hazard types

Dataset D:
Post-disaster street-view images



It consists of post-disaster ground-level images, including geo-referenced wildfire inspections (D1) and multi-hazard web-sourced scenes without geolocation (D2).

a. Events

- 9 sites, 4 continents: USA ×4, Haiti, Morocco, Libya, Turkey, Japan
- Earthquake, flood, hurricane, wildfire (+ ice storm, drought in D2)

b. Key statistics

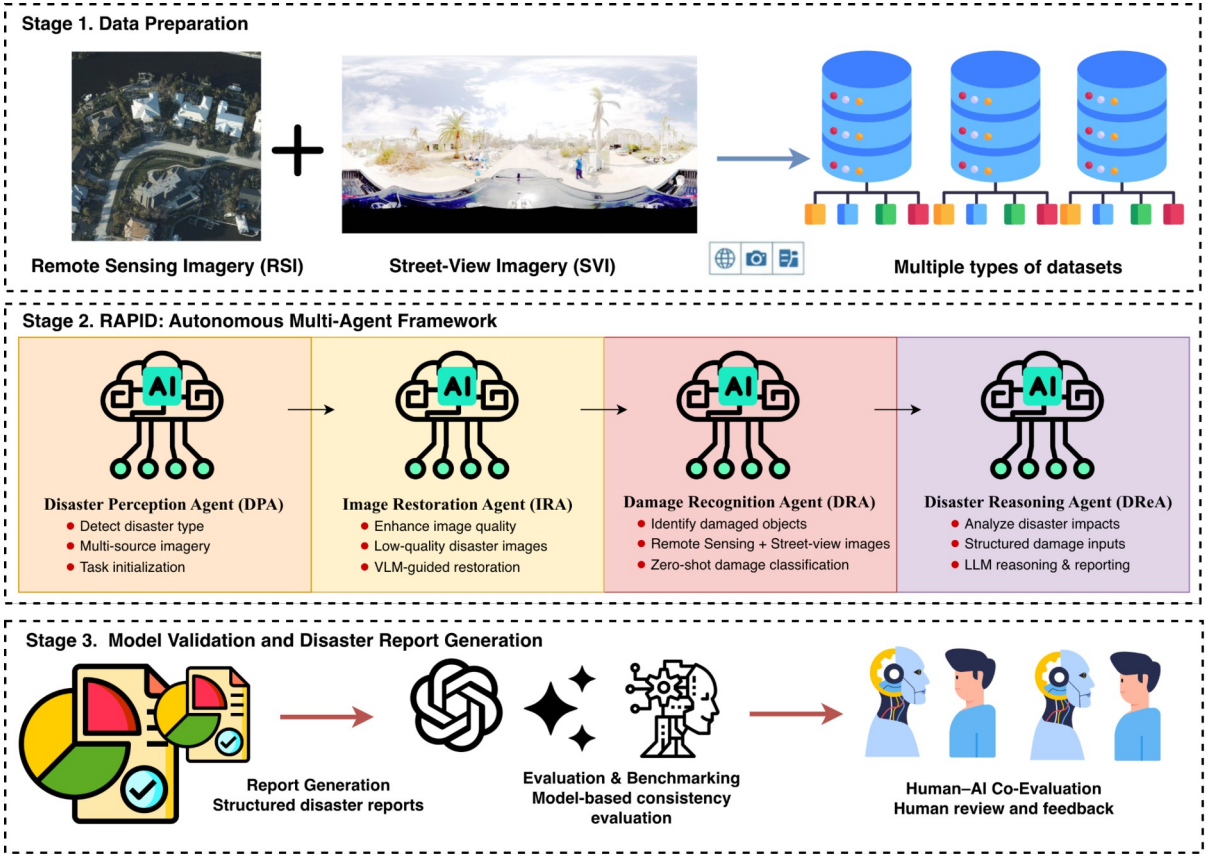
- 4 datasets, ~1,100 images (A 150 · B 150 · C 350 · D 471)
- Balanced severity in A / B; official 0–4 scale in D1

c. Datasets

- A: post-event street view + satellite (CVDisaster)
- B: pre/post street view (BiTemporal StreetView)
- C: pre/post satellite + building masks (BRIGHT)
- D: post-event street view, geo-referenced or web-sourced

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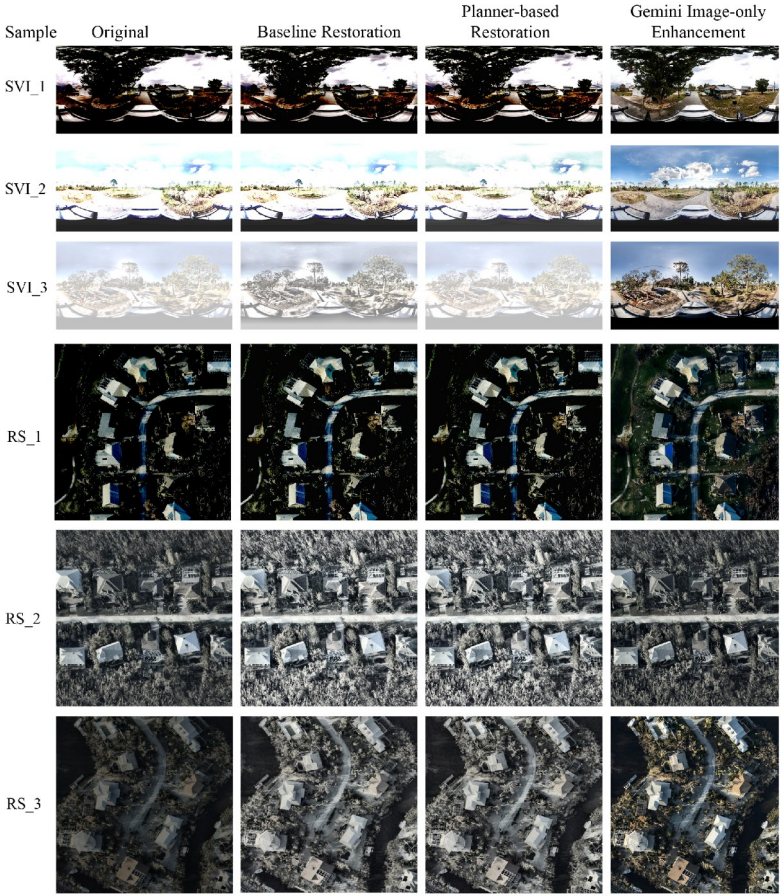
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

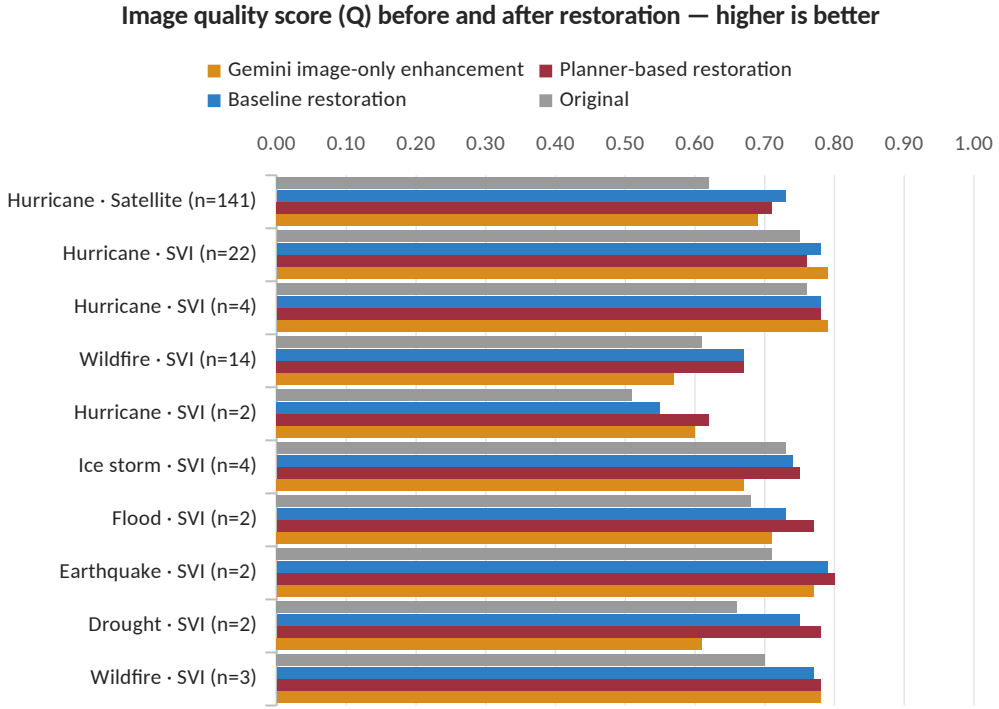


Method:

- **Cross-view imagery input**
- **Zero-shot multi-agent assessment**
- **Structured report generation**

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

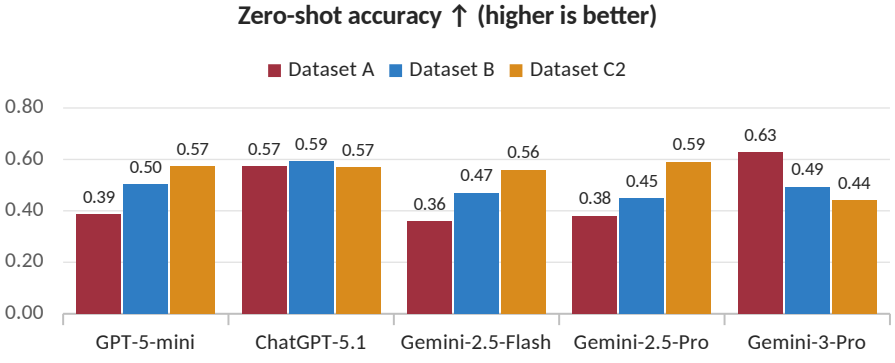


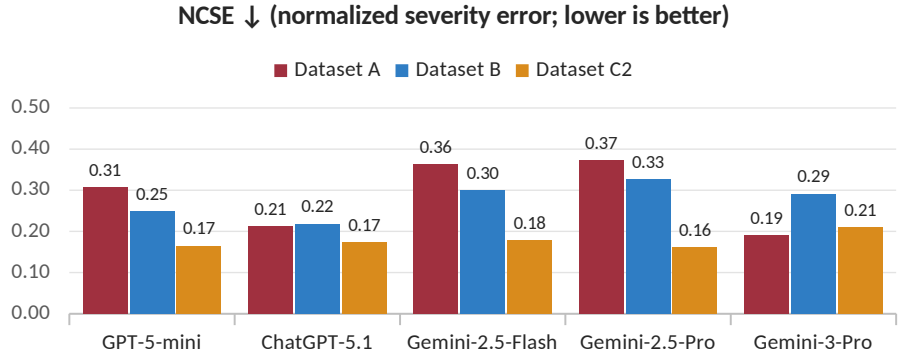


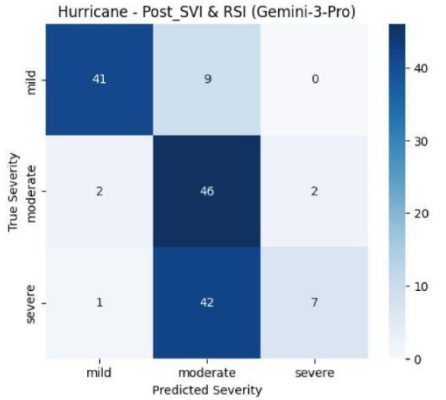
Baseline and planner-based restoration raise Q in every row (e.g., 0.51 → 0.62, 0.71 → 0.80); Gemini image-only enhancement helps hurricanes but lowers Q for wildfire, ice storm, and drought.

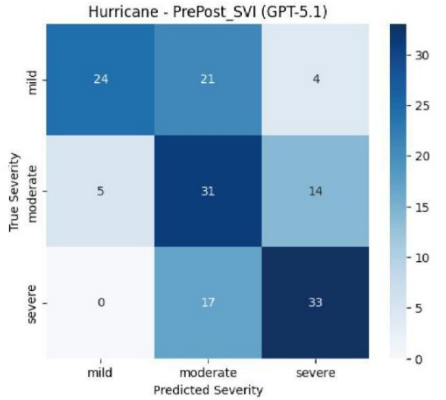
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

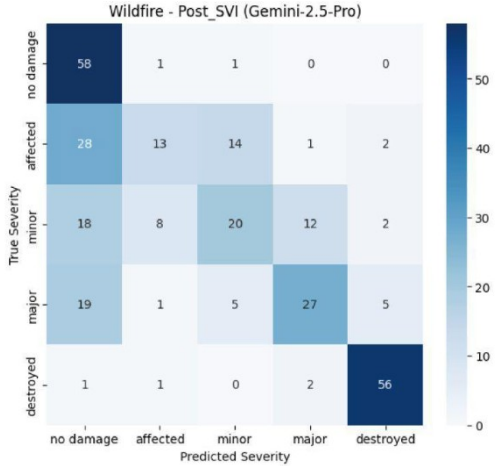
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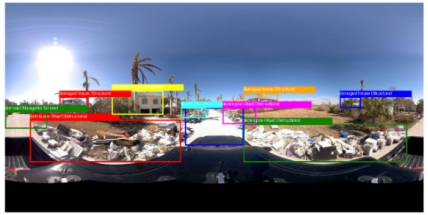




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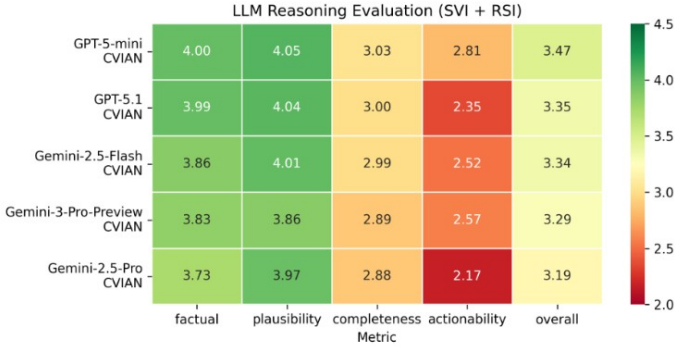
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

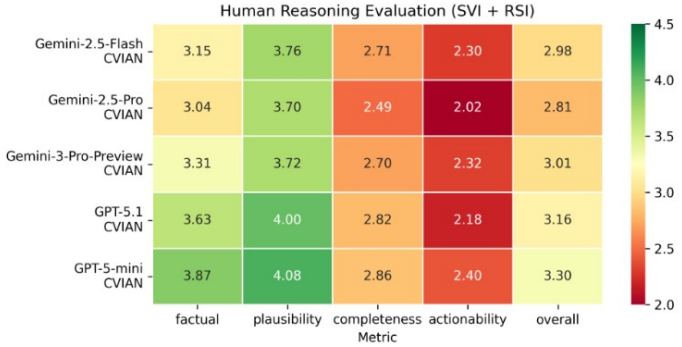
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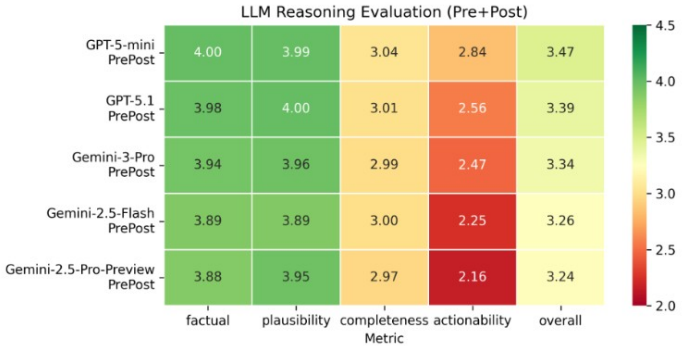
Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
 	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**			

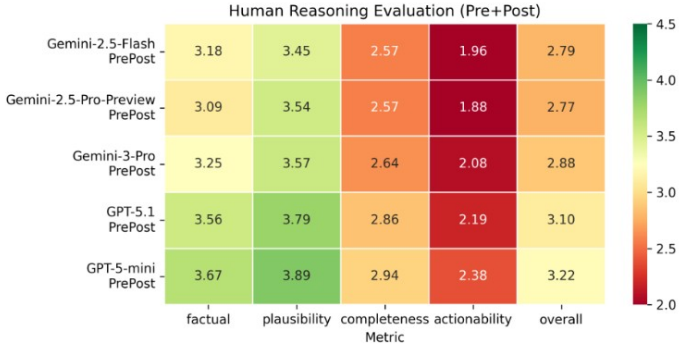
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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery









Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted by ACM SIGSPATIAL)

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RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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Abstract

Due to the increasing frequency and intensity of climate extreme events, there is a clear demand for intelligent, scalable, and autonomous approaches to disaster damage assessment. Existing methods, largely based on supervised learning and task-specific fine-tuning strategies, struggle to generalize under domain shifts, long-tailed data distributions, and heterogeneous geospatial data sources, especially in disaster scenarios. In addition, they often lack the ability to integrate and reason effectively across multimodal geospatial information, such as satellite images and street-view images. In this paper, we introduce RAPID, a reproducible multi-agent pipeline for interpretable disaster damage assessment, encompassing the assessment of damage levels, interpretation of damage types and degrees, and generation of actionable suggestions for response, remediation, and recovery. RAPID coordinates multiple specialized agents to perform cross-view understanding, image restoration, structured damage recognition, and geographical reasoning across heterogeneous data modalities. Without task-specific fine-tuning, RAPID supports zero-shot damage assessment by jointly leveraging complementary information from remote sensing and ground-level perspectives. Furthermore, the system produces fine-grained, interpretable damage assessments and automatically generates location-specific, decision-relevant disaster reports to support early-stage emergency response. We evaluated and demonstrated the performance of the proposed RAPID framework across multiple disaster using different sets of cross-view imagery inputs, including street-view images before and after the disaster, remote sensing imagery after the disaster, and street-view image pairs. Experiments show that RAPID achieves an overall accuracy of 0.92 for multi-disaster type classification and up to 0.627 for cross-view damage severity prediction, highlighting its potential as a foundational framework for autonomous disaster intelligence.

CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → *Artificial intelligence*; *Computer vision tasks.*

Keywords

Disaster damage assessment, vision-language models, cross-view imagery, zero-shot learning, multi-agent pipeline

ACM Reference Format:

Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. In *The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26), November 03–06, 2026, Riverside, CA, USA.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3841645.3843346

Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. In The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26), November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3841645.3843346

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STAGE 3 · OBJECTIVE 3

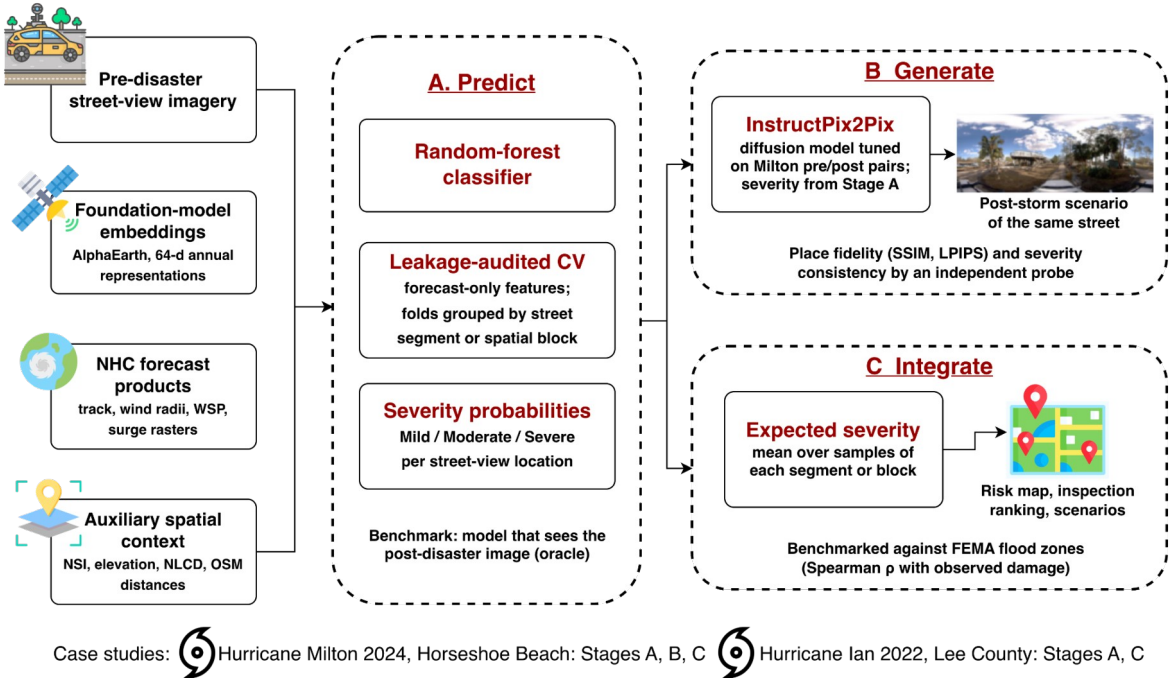
Prepare and deliver better

Challenge 3 · Assessment alone does not close the resilience loop

Build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters.

Case 6	Seeing damage before it happens: anticipatory street-level prediction	Ongoing
Case 7	RAY: an accountable GeoAI system for place-based disaster intelligence	Ongoing

Case 6: Seeing Disaster Damage Before It Happens: Predicting and Visualizing Street-Level Impacts from Multimodal Pre-Event Information (Ongoing)



Case studies: Hurricane Milton 2024, Horseshoe Beach: Stages A, B, C Hurricane Ian 2022, Lee County: Stages A, C

Research Question:

- **Can we anticipate post-disaster street-level damage using pre-disaster multi-modal data, Earth observation embeddings, and hazard forecasts?**

Objective:

- **To develop a preparedness-oriented GeoAI framework that transforms pre-disaster data and forecasts into anticipatory damage predictions, spatial risk prioritization, and scenario visualizations.**

Case 6 · What I will do (Ongoing, ~20% complete)

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Predict

- Random-forest classifier on pre-event street view, foundation-model embeddings, forecasts, and spatial context
- Severity probability per street location
- Leakage-audited CV grouped by street segment

Generate

- Instruction-tuned diffusion model (InstructPix2Pix) trained on Milton before/after pairs
- A visual scenario of the expected damage for each location
- Checked for place fidelity (SSIM, LPIPS) and severity consistency

Integrate

- Risk map: expected severity + scenario per block
- Prioritizes inspection routes before landfall
- Benchmarked against FEMA flood zones (pre-event standard)

Data and study areas

- Hurricane Milton 2024, Horseshoe Beach FL: train and test (Predict, Generate, Integrate)
- Hurricane Ian 2022, Lee County FL: cross-event transfer test (Predict, Integrate)

Deliverables and milestones

- Fall 2026: prediction + transfer results, scenario generator
- Spring 2027: risk maps vs. FEMA benchmark, manuscript draft
- Summer 2027: dissertation chapter

Expected insight: where to inspect first, before the storm arrives.

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Case 7: RAY: Resilience Assistant for You
An Accountable GeoAI System for Place-Based Disaster Intelligence (Ongoing)

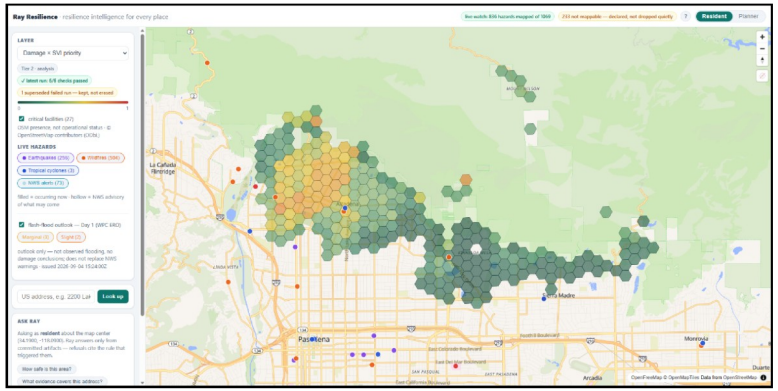
Data plane

GitHub Actions, hourly + on push
- Tier-1 watch: USGS · NWS · NHC · NIFC
- Deep-case builders: Eaton, Milton, Ian
- Every artifact: SHA-256 + audit row
- Append-only snapshots

Agent plane (Ray)

Any OpenAI-compatible endpoint
pre-check → grounded generation → citation post-check → audit
- Cites |artifact:...| or refuses
- Every refusal names its rule

Presentation plane



Steward Harness

CI gate: the assembled public site is verified against the distribution plane; a violating deploy fails

Outcome validity
Executable spatial checks: CRS, join integrity, bounds. Per-tile uncertainty is mandatory, or the build fails.

Process validity
SHA-256 per artifact; append-only audit rows (agent, inputs, UTC). Failures recorded, never erased.

Institutional validity
One YAML policy, two planes: claim = what may be asserted; distribution = what may ship. Ordered rules, default deny.

Verifiability
retained > re-derivable > cited-only; weakest link. A license attribute gates redistribution by rule.

Research Question:

How can a GeoAI agent analyze disasters on its own, yet have every answer authorized, evidence-faithful, and reproducible, enforced by the system rather than promised by the model?

Objective:

Build RAY: an agent wrapped in a Steward Harness (policy check → grounded analysis → citation check → audit & replay), delivered through a WebGIS / mobile console for planners and residents.



Link to the 3-minute demo video:
https://youtu.be/qn7u9lRuNUM

Case 7 · Plan and evaluation of RAY (Ongoing, ~40% complete)

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Build

- Agent plane on open, OpenAI-compatible models with spatial tools and live data (USGS, NWS, NOAA, NHC, Census)
- Steward Harness: policy check → grounded analysis → citation check → audit & replay
- WebGIS and mobile console for planners and residents

Evaluate on four validities

- Outcome: spatial checks pass (CRS, joins, bounds); uncertainty reported
- Process: every step logged and replayable
- Institutional: answers obey policy; refusals name the rule
- Verifiability: every sentence cites an inspectable artifact

Tests

- Question bank across the four resilience phases, with known answers
- With vs. without the harness: answer accuracy, citation faithfulness, correct refusals
- Review sessions with planners and residents

Milestones

- Done: data plane, agent plane, console prototype, demo
- Fall 2026: harness gates complete, question bank
- Spring 2027: evaluation + user review, manuscript
- Summer 2027: dissertation chapter and final defense

Expected insight: trustworthy, place-based answers that a planner can act on and audit.

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Publications and Awards

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As of October 2026 · Google Scholar: 434 citations, h-index 8, i10-index 7.

PUBLICATIONS

8 first-authored publications during Ph.D.

- 2 journal articles (CEUS 2025; Natural Hazards under review)
- 5 conference proceedings (ACM SIGSPATIAL 2026; ACM SIGSPATIAL Workshop GeoSearch '26; IEEE IGARSS 2026; CaGIS 2026; ICC 2025)
- 1 chapter under review (CRC Press)

14 co-authored publications

- 8 journal articles (e.g., International Journal of Geographical Information Science, Transactions in GIS, Int. J. Applied Earth Observation and Geoinformation, Urban Informatics, Journal of Transport Geography, etc.)
- 3 conference papers (ACM SIGSPATIAL 2026 short paper; GeoAI 2026; AGU 2025) and 3 preprints



HONORS AND AWARDS

A total of 15 honors received during Ph.D.

2026

- 2nd Place, AAG GISSG Student Honors Paper Competition
- AAG International Geographic Information Fund (IGIF) Scholarship
- CaGIS Doctoral Scholarship Award
- IGARSS 2026 Travel Grant
- NSF I-GUIDE Summer School Travel Award
- ACM SIGSPATIAL 2026 Travel Award
- Geography Graduate Student Enhancement Scholarship, Department of Geography, etc.

2025

- Best Student Paper Award, International Cartographic Conference (ICC 2025)
- Environment and Sustainability Graduate Fellow Award, TAMU
- TAMIDS Student Ambassador Scholarship
- CaGIS International Travel Grant, etc.

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Research Schedule

Now

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Tasks	Fall 2024	Spring 2025	Summer 2025	Fall 2025	Spring 2026	Summer 2026	Fall 2026	Spring 2027	Summer 2027
Case Study 1	Done								
Case Study 2		Done							
Case Study 3				Done					
Case Study 4					Done				
Case Study 5					Done				
Case Study 6							In Progress (20%)		
Case Study 7							In Progress (40%)		
Prelim Exam					Done				
Proposal Defense							In Progress		
Dissertation Writing							In Progress		
Final Defense & Graduation									Planned

Done In Progress Planned

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Q&A

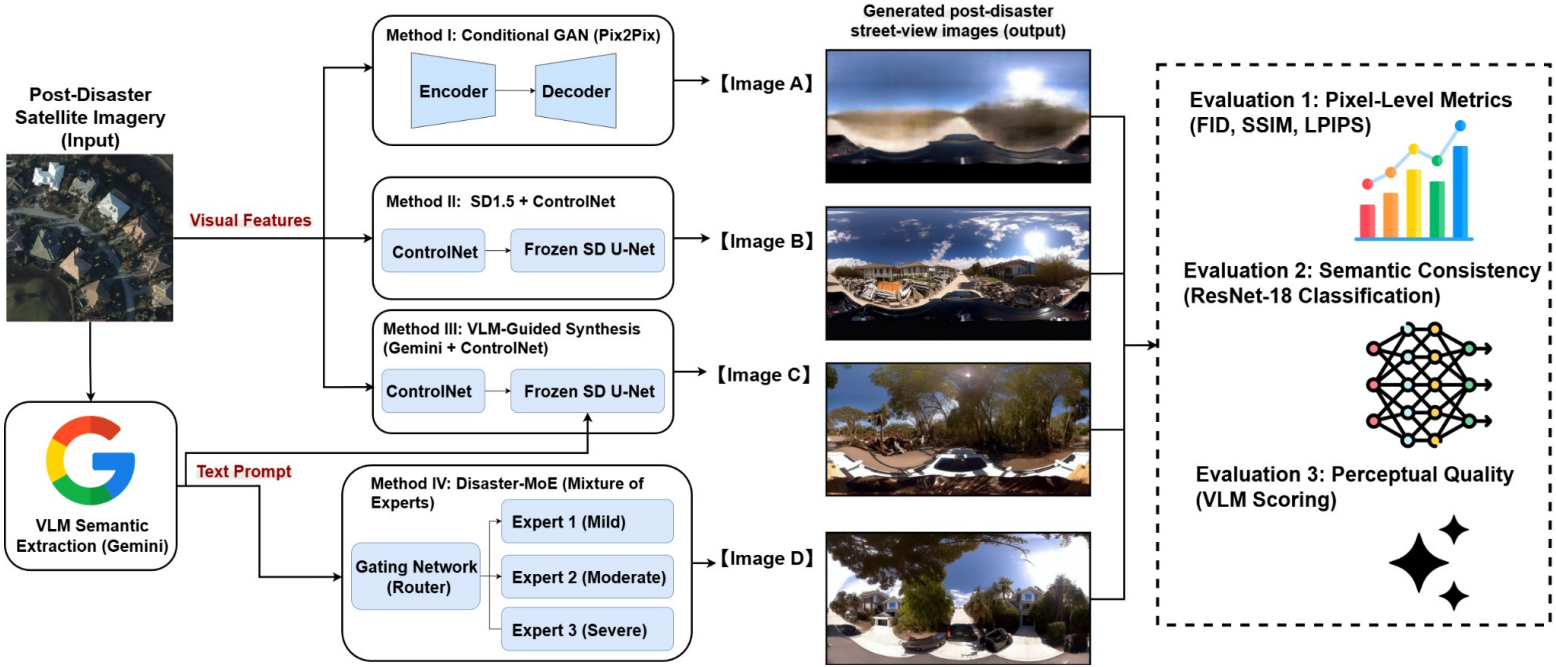
Welcome any comments/questions

yyang295@tamu.edu

See better · Respond faster · Prepare and deliver better

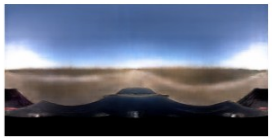
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Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)



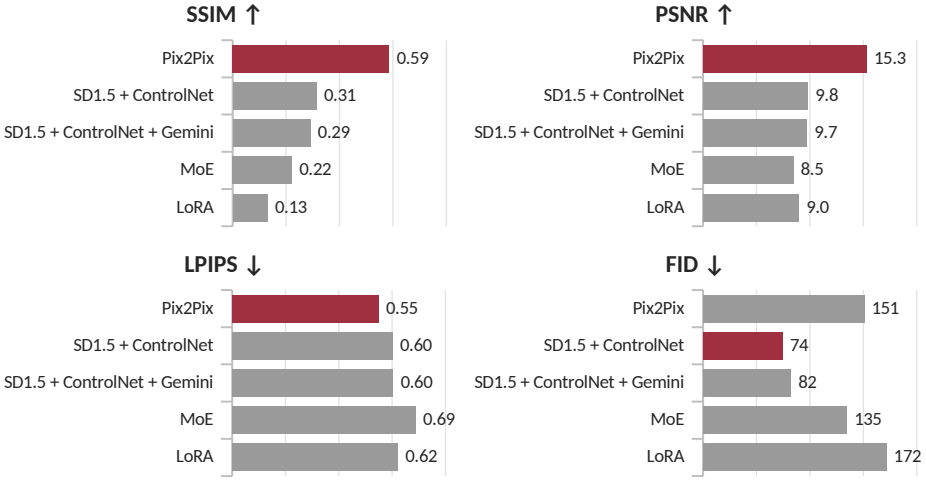
RQ1: Can generative vision models synthesize realistic post-disaster street views from satellite imagery?
RQ2: Do explicit disaster semantics improve synthesis fidelity?
RQ3: How can we reliably evaluate cross-view disaster synthesis beyond pixel metrics?

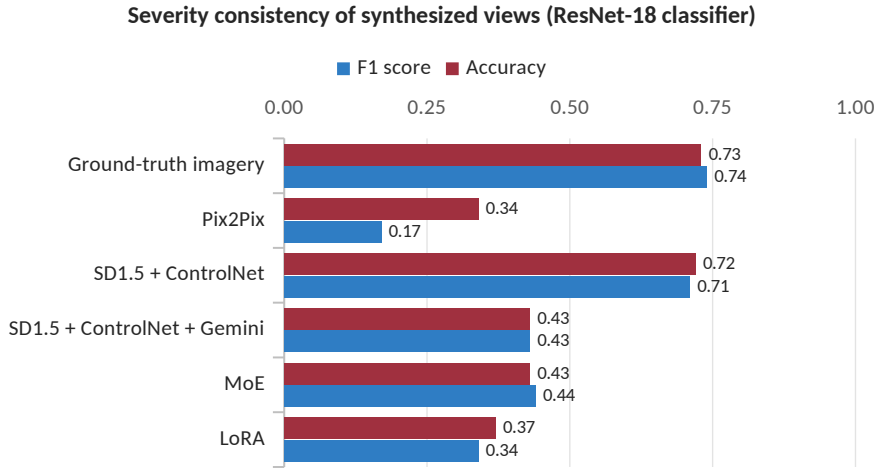
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	Satellite	Pix2Pix	SD1.5+ControlNet	SD1.5+ControlNet+Gemini	MoE	Ground Truth
Mild						
Moderate						
Severe						

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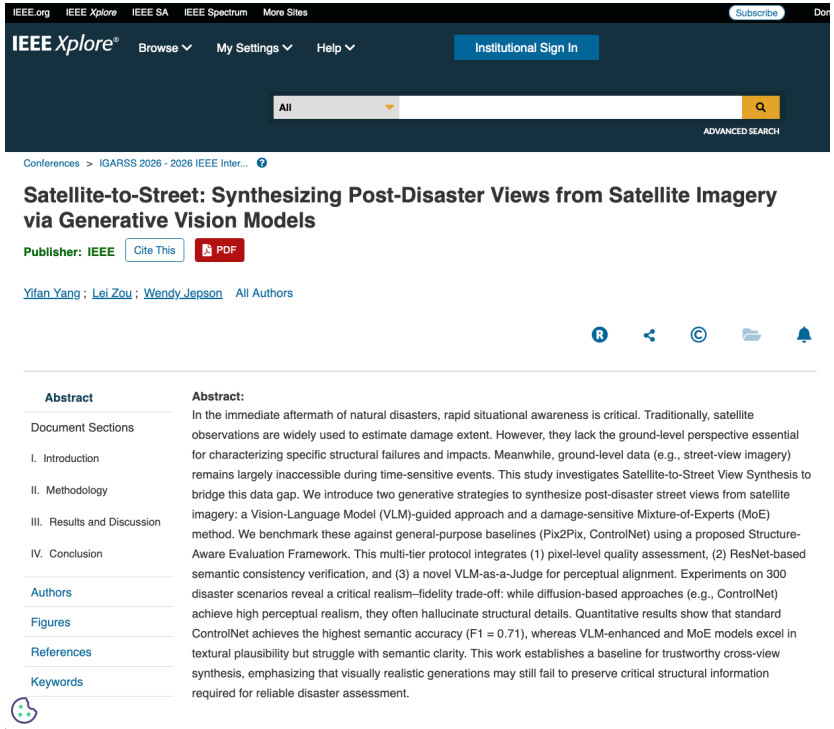


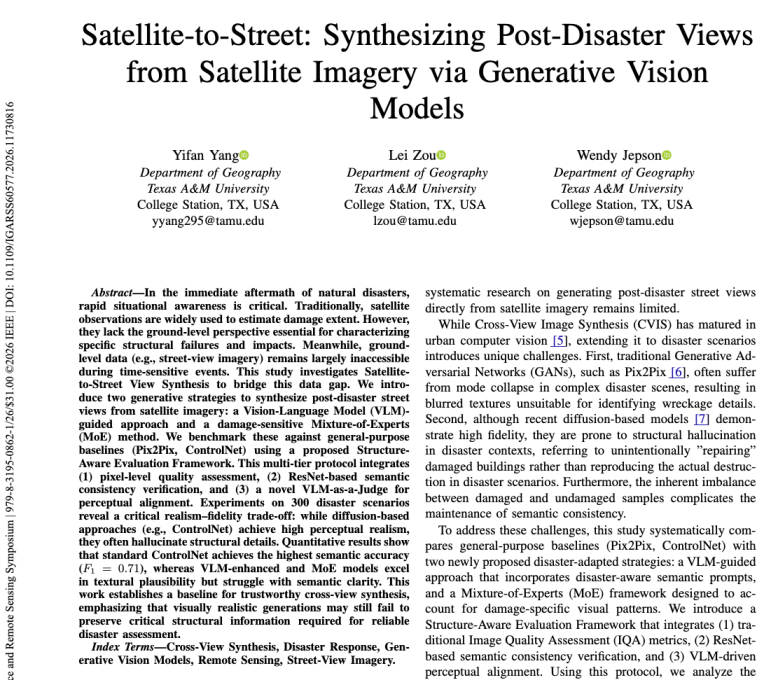


Pixel-level and perceptual image-quality metrics (best in maroon)

- Single-model approaches can't yet balance visual plausibility with strict structural alignment
- High perceptual realism ≠ high semantic accuracy — evaluation must be structure-aware
- Trustworthiness, not fidelity alone, should be the target for disaster generation

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Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models

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Abstract*—In the immediate aftermath of natural disasters, rapid situational awareness is critical. Traditionally, satellite observations are widely used to estimate damage extent. However, they lack the ground-level perspective essential for characterizing specific structural failures and impacts. Meanwhile, ground-level data (e.g., street-view imagery) remains largely inaccessible during time-sensitive events. This study investigates Satellite-to-Street View Synthesis to bridge this data gap. We introduce two generative strategies to synthesize post-disaster street views from satellite imagery: a Vision-Language Model (VLM)-guided approach and a damage-sensitive Mixture-of-Experts (MoE) method. We benchmark these against general-purpose baselines (Pix2Pix, ControlNet) using a proposed Structure-Aware Evaluation Framework. This multi-tier protocol integrates (1) pixel-level quality assessment, (2) ResNet-based semantic consistency verification, and (3) a novel VLM-as-a-Judge for perceptual alignment. Experiments on 300 disaster scenarios reveal a critical realism–fidelity trade-off: while diffusion-based approaches (e.g., ControlNet) achieve high perceptual realism, they often hallucinate structural details. Quantitative results show that standard ControlNet achieves the highest semantic accuracy ($F_1 = 0.71$), whereas VLM-enhanced and MoE models excel in textural plausibility but struggle with semantic clarity. This work establishes a baseline for trustworthy cross-view synthesis, emphasizing that visually realistic generations may still fail to preserve critical structural information required for reliable disaster assessment.

***Index Terms*—Cross-View Synthesis, Disaster Response, Generative Vision Models, Remote Sensing, Street-View Imagery.**

systematic research on generating post-disaster street views directly from satellite imagery remains limited.

While Cross-View Image Synthesis (CVIS) has matured in urban computer vision [5], extending it to disaster scenarios introduces unique challenges. First, traditional Generative Adversarial Networks (GANs), such as Pix2Pix [6], often suffer from mode collapse in complex disaster scenes, resulting in blurred textures unsuitable for identifying wreckage details. Second, although recent diffusion-based models [7] demonstrate high fidelity, they are prone to structural hallucination in disaster contexts, referring to unintentionally "repairing" damaged buildings rather than reproducing the actual destruction in disaster scenarios. Furthermore, the inherent imbalance between damaged and undamaged samples complicates the maintenance of semantic consistency.

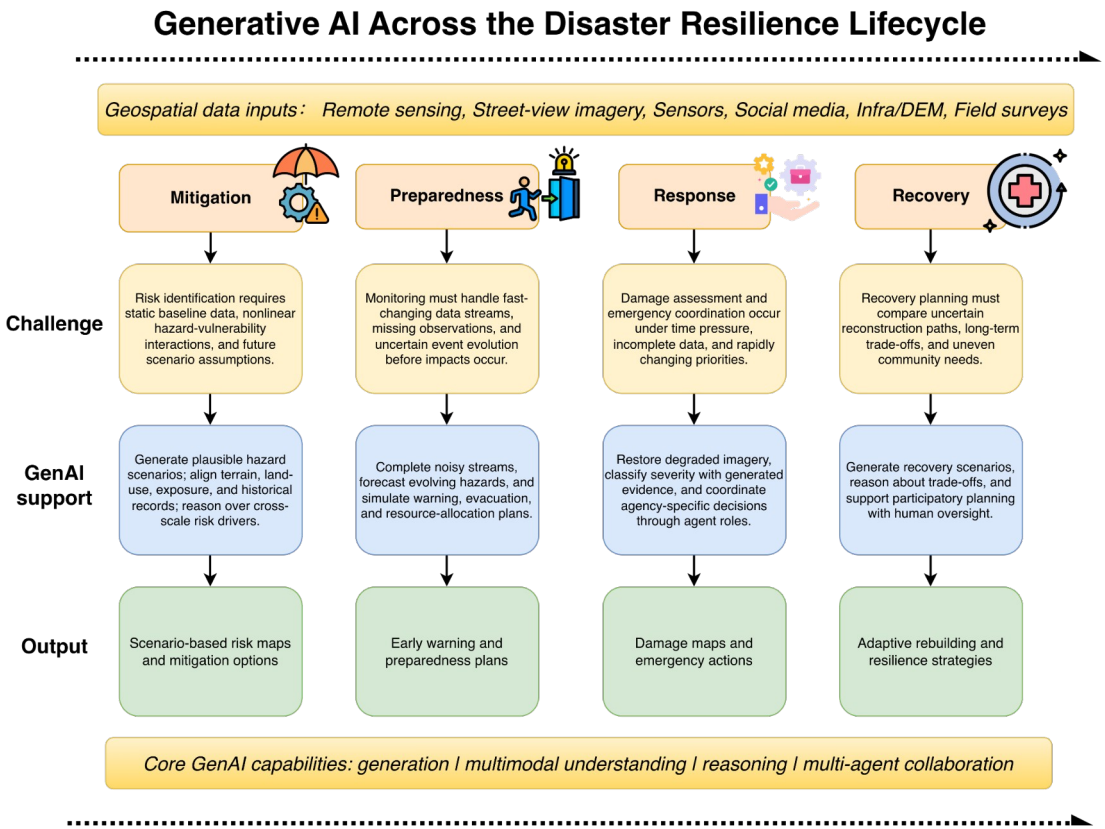
To address these challenges, this study systematically compares general-purpose baselines (Pix2Pix, ControlNet) with two newly proposed disaster-adapted strategies: a VLM-guided approach that incorporates disaster-aware semantic prompts, and a Mixture-of-Experts (MoE) framework designed to account for damage-specific visual patterns. We introduce a Structure-Aware Evaluation Framework that integrates (1) traditional Image Quality Assessment (IQA) metrics, (2) ResNet-based semantic consistency verification, and (3) VLM-driven perceptual alignment. Using this protocol, we analyze the

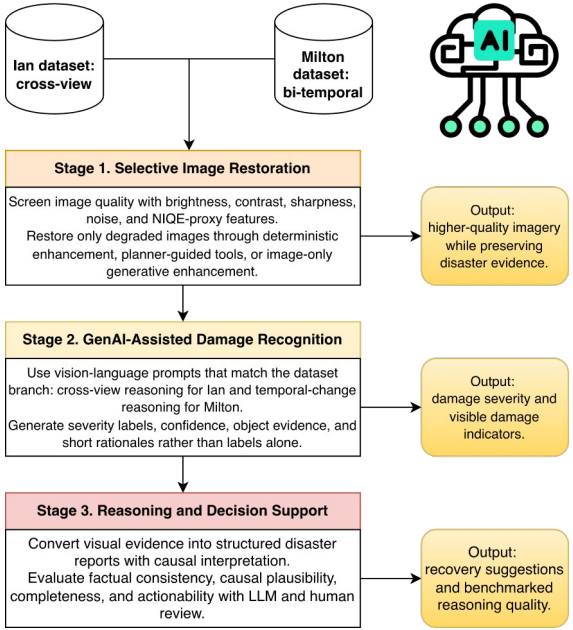
ze and Remote Sensing Symposium | 979-8-3195-0862-1/26$31.00 ©2026 IEEE | DOI: 10.1109/IGARSS60577.2026.11730816

Y. Yang, L. Zou and W. Jepson, "Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models," IGARSS 2026 - 2026 IEEE International Geoscience and Remote Sensing Symposium, Washington DC, DC, USA, 2026, pp. 1764–1768, doi: 10.1109/IGARSS60577.2026.11730816.

Case 5: Generative AI for Disaster Resilience

(Under Review, Geography in the Age of Generative AI (CRC Press))





Case 8: When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment (Accepted by the 5th ACM SIGSPATIAL GeoSearch '26)



MISSION

The proliferation of geospatial data has accelerated due to advances in Earth Observation (EO) technologies, the widespread deployment of Internet of Things (IoT) sensors, the rise of open science initiatives, and the increasing availability of cloud-native and AI-driven geospatial platforms. This data is now highly heterogeneous in format, modality, resolution, and temporal scale, and is being generated at unprecedented speed and volume. These trends introduce significant challenges—and opportunities—in how to efficiently search, integrate, and mine massive, multimodal geospatial datasets to reveal meaningful patterns and support informed decision-making.

Example topics include but are not limited to:

- Search and Retrieval using LLMs, VLMs and World Model
- Geospatial Reasoning for Effective Search and Retrieval
- Spatial Representation Learning for Search and Retrieval
- Spatial Pattern Matching
- Geospatial Knowledge Graphs for Search
- Multimedia Information Retrieval

When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment

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Abstract

Satellite and street-view imagery see different parts of a damaged building: roofs from above, facades and debris from the ground. Average accuracy hides whether fusion methods use this complementary evidence. We study the *conflict cases*, the samples on which independently trained single-view models disagree, as a diagnostic subset. On paired collections from the 2025 Eaton wildfire and Hurricanes Ian and Milton, conflicts make up 9.6–33.2% of the test data. Fusion improves conflict accuracy over either view, yet an oracle that selects the correct single view per conflict remains 0.19–0.32 above the best evaluated fusion method. A linear fusion gate over building-visibility features, calibrated confidences, and disagreement recovers part of this headroom where ground photographs are aimed at the building: on Eaton it beats end-to-end fusion by +0.072 conflict accuracy ($p < 10^{-4}$) and calibrated averaging by +0.051 ($p = 0.0001$); on the panoramic hurricane datasets it is indistinguishable from end-to-end fusion. Cropping panoramas toward the building raises the conflict gain of fusion, whereas random crops of the same size do not. The spatial density of conflicts, computed without labels, correlates with tile-level wildfire damage (Spearman $r = 0.615$, 25 tiles). Conflict cases thus show how much complementary evidence fusion uses, which view carries it under each hazard and capture setting, and where current methods fall short.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems.*

Collections of Geospatial Data (GeoSearch '26), November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3849732.3857333

1 Introduction

Building-level damage assessment after a disaster increasingly draws on two image collections at once. Satellite imagery covers a whole region within hours and is the basis of standardized damage mapping [2], but it mainly shows roofs. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot fully detect [11, 13]. The views are complementary: a facade can be gutted under an unbroken roof. Methods that pair them serve geolocalization and damage mapping [5, 6, 12], multimodal arbitration [10], and agentic assessment pipelines [8, 9], and usually report that fusion improves average accuracy over either view alone.

Average accuracy says little about whether this evidence is used, because for most buildings the two views agree. The informative records are the *conflict cases*, the samples on which independently trained single-view models disagree: at least one view carries evidence the other lacks, whether damage is visible only from one side, vegetation hides the building, or the views are misaligned. Conflicts are thus a focused subset for asking how much complementary evidence fusion recovers, which view carries it under a given hazard and capture setting, and how the views can be combined.

This study examines cases in which damage assessments derived from satellite imagery conflict with those derived from street-view